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Alkylation of enolates

Connections

➡ Building on

- Enols and enolates **ch20**
- Electrophilic addition to alkenes **ch19**
- Nucleophilic substitution reactions **ch15**
- Conjugate additions **ch22**

Arriving at

- How to make new C–C bonds using carbonyl compounds as nucleophiles
- How to prevent carbonyl compounds reacting with themselves

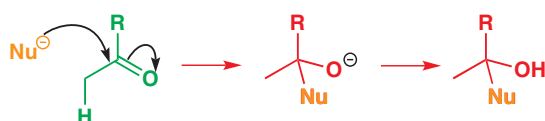
➡ Looking forward to

- Forming C–C bonds by reacting nucleophilic enolates with electrophilic carbonyl compounds **ch26**
- Retrosynthetic analysis **ch28**

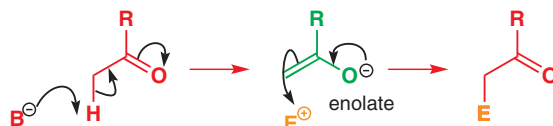
Carbonyl groups show diverse reactivity

In earlier chapters we discussed the two types of reactivity displayed by the carbonyl group. We first described reactions that involve nucleophilic attack on the carbon of the carbonyl, and in Chapter 9 we showed you that these are among the best ways of making new C–C bonds. In this chapter we shall again be making new C–C bonds, but using *electrophilic* attack on carbonyl compounds: in other words, the carbonyl compound will be reacting as the nucleophile in the reaction. We introduced the nucleophilic forms of carbonyl compounds—enols and enolates—in Chapter 20. There you saw them reacting with electrophiles based on elements other than carbon, but they will also react well with carbon electrophiles provided the reaction is thoughtfully devised. Much of this chapter will concern that phrase, ‘thoughtfully devised’.

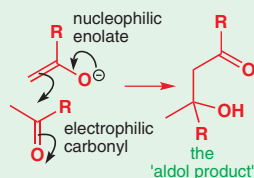
carbonyl compound acts as an electrophile



enolate acts as a nucleophile



■ In the next chapter we shall talk about how to promote and control the reactions of carbonyl compounds with themselves, known as *aldol reactions*.



Thought is needed to ensure that the carbonyl compound exhibits the right sort of reactivity. In particular, the carbonyl compound must not act as an electrophile when it is intended to be a nucleophile. If it does, it may react with itself to give some sort of dimer—or even a polymer—rather than neatly attacking the desired electrophile. This chapter will consider ways of avoiding unwanted nucleophilic attack at the carbonyl C=O bond.

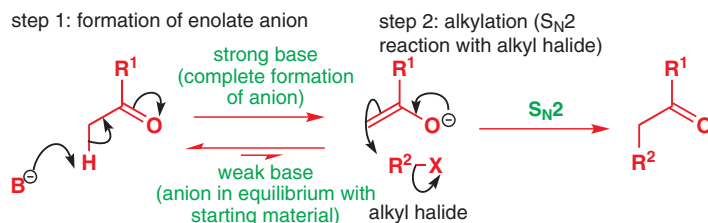
Fortunately, over the last four decades lots of thought has *already* gone into the problem of controlling the reactions of enolates with carbon electrophiles. This means that there are many excellent solutions to the problem: our task in this chapter is to help you understand which to use, and when to use them, in order to design useful reactions.

Some important considerations that affect all alkylations

The alkylations in this chapter will each consist of two steps. The first is the formation of a stabilized anion—usually (but not always) an enolate—by deprotonation with base.

Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

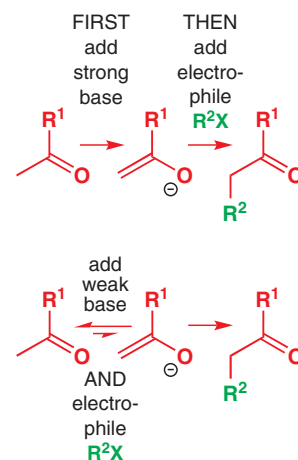
The second is a substitution reaction: attack of the nucleophilic anion on an electrophilic alkyl halide. All the factors controlling S_N1 and S_N2 reactions, which we discussed at length in Chapter 15, are applicable here.



In each case, we shall take one of two approaches to the choice of base.

- A strong base (with a conjugate acid of pK_a greater than that of the carbonyl compound) can be chosen to *deprotonate the starting material completely*. There is complete conversion of the starting material to the anion before addition of the electrophile, which is added in a subsequent step.
- Alternatively, a weaker base may be used *in the presence of the electrophile*. The weaker base will not deprotonate the starting material completely because its conjugate acid has a lower pK_a than the carbonyl compound: only a small amount of anion will be formed, but that small amount will react with the electrophile. More anion is formed as alkylation uses it up.

The second approach is easier practically (just mix the starting material, base, and electrophile), but works only if the base and the electrophile are compatible and don't react together. With the first approach, which is practically more demanding, the electrophile and base never meet each other, so their compatibility is not a concern. We shall start with some compounds that avoid the problem of competing aldol reactions completely because they are not electrophilic enough to react with their own nucleophilic derivatives.

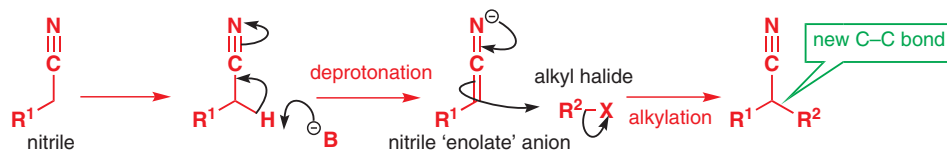


Nitriles and nitroalkanes can be alkylated

Problems that arise from the electrophilicity of the carbonyl group can be avoided by replacing $C=O$ by functional groups that are much less electrophilic but are still able to stabilize an adjacent anion. We shall consider two examples, both of which you met in Chapter 20.

Alkylation of nitriles

The nitrile group, which mirrors the carbonyl group in general reactivity, is much less easily attacked by nucleophiles (N is less electronegative than O). The anion formed by deprotonating a nitrile using strong base will not react with other molecules of nitrile but will react very efficiently with alkyl halides. The slim, linear structure of the anions makes them good nucleophiles for S_N2 reactions.



The nitrile does not have to be deprotonated completely for alkylation: with sodium hydroxide only a small amount of anion is formed. In the example below, such an anion reacts with propyl bromide to give 2-phenylpentanenitrile.



➔ You met nitrile hydrolysis and addition reactions, for example, in Chapter 10.

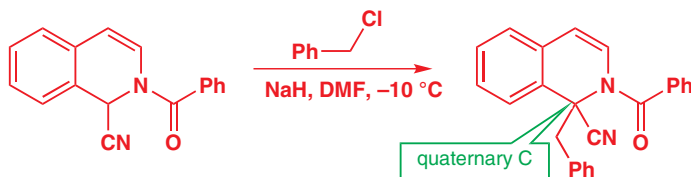
■ The pK_a of acetonitrile (MeCN) is 25.

Phase transfer catalysis

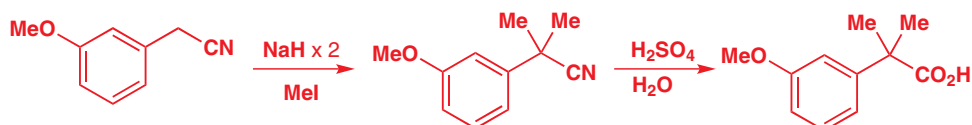
This reaction is carried out in a two-phase mixture (water + an immiscible organic solvent) to prevent the hydroxide and propyl bromide reacting together to give propanol. The hydroxide stays in the aqueous layer, and the other reagents stay in the organic layer. A tetraalkylammonium chloride (benzyltriethylammonium chloride $BnEt_3N^+Cl^-$) is needed as a *phase transfer catalyst* to allow sufficient hydroxide to enter the organic layer to deprotonate the nitrile.

Nitrile-stabilized anions are so nucleophilic that they will react with alkyl halides rather well even when a crowded quaternary centre (a carbon bearing no H atoms) is being formed. In this example the strong base, sodium hydride, was used to deprotonate the branched nitrile completely and benzyl chloride was the electrophile. The greater reactivity of benzylic electrophiles compensates for the poorer leaving group. In DMF, the anion is particularly reactive because it is not solvated (as you saw in Chapter 12, p. 255, DMF solvates only the Na^+ cation).

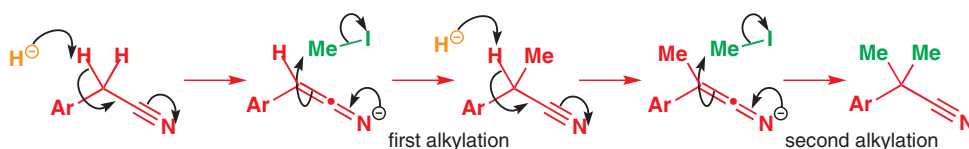
Remember our discussion about the lack of nucleophilicity of hydride (H^-) in Chapter 6? Here is hydride acting as a base even in the presence of the electrophile: there was no need to do this reaction in two steps because the base and electrophile don't react with one another.



The compatibility of sodium hydride with electrophiles means that, by adding two equivalents of base, alkylation can be encouraged to occur more than once. This dimethylated acid was required in the synthesis of a potential drug, and it was made in two steps from a nitrile.



The '•' in the second and fourth structures indicates the linear carbon atom, which might get overlooked if it were not indicated in this way.



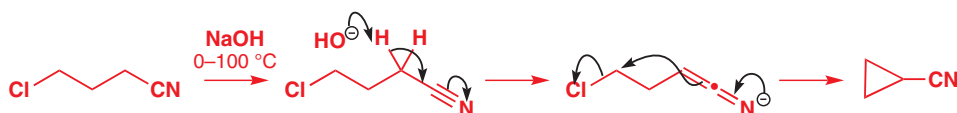
Multiple alkylation

Multiple alkylation is not always desirable, and one of the side reactions in alkylations that are intended to go only once is the formation of doubly alkylated products. These arise when the first alkylation product still has acidic protons and can be deprotonated to form another anion, which may in turn react further. Clearly, this is more likely to be a problem if the base is present in excess and can usually be restricted by using only one equivalent of the electrophile.

With two nitrile groups, the delocalized anion is so stable that even a weak, neutral amine (triethylamine) is sufficiently basic to deprotonate the starting material. Here double alkylation again takes place, in 100% yield: note that the electrophile is good at $\text{S}_{\text{N}}2$, and the dipolar aprotic solvent DMSO (like DMF) cannot solvate the 'enolate' anion, making it more reactive.



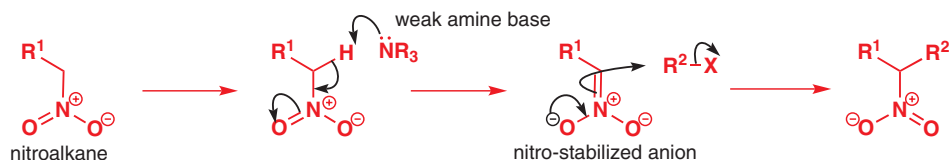
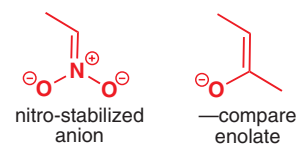
If the electrophile and the nitrile are in the same molecule and the spacing between them is appropriate, then intramolecular alkylation entails cyclization to form rings. The preparation of a cyclopropane is shown using sodium hydroxide as the base and chloride as a leaving group. With an intramolecular alkylation, the base and the electrophile necessarily have to be present together, but the cyclization is so fast that competing $\text{S}_{\text{N}}2$ substitution of Cl^- by HO^- is not a problem.



Alkylation of nitroalkanes

The powerful electron-withdrawing nature of the nitro group means that deprotonation is possible even with quite weak bases. The pK_{a} of MeNO_2 is 10, about the same as phenol.

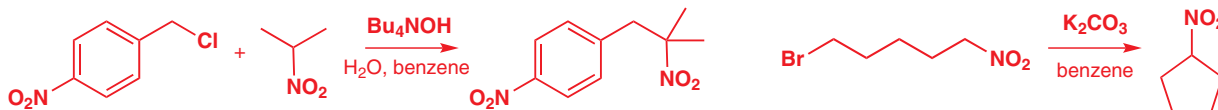
Protons adjacent to a nitro group are in fact about as acidic as the same proton adjacent to two carbonyl groups; you can think of a nitro group as having double the electron-withdrawing power of a carbonyl group. Nitro-stabilized anions ('nitronate anions') react with carbon electrophiles and a wide variety of nitro-containing products can be produced. The anions are not, of course, enolates, but replacing the nitrogen with a carbon should help you to recognize the close similarity of these alkylations with the enolate alkylations described later.



Surprisingly few simple nitroalkanes are commercially available but more complex examples can be prepared readily by alkylation of the anions derived from nitromethane, nitroethane, and 2-nitropropane. For example, deprotonation of nitropropane with butyllithium followed by the addition of butyl iodide gives 3-nitroheptane in good yield. This reaction really does have to be done in two steps: BuLi is not compatible with alkyl halides!



Nitroalkanes can be alkylated in a single step with hydroxide as a base: phase transfer conditions (see p. 585) keep the HO⁻ and the electrophile apart, preventing alcohol formation. The reaction below on the left works despite the quaternary carbon atom in the product. The reaction on the right gives a cyclic nitroalkane: now there really is no alternative: the base and electrophile must cohabit in the reaction mixture, so a weaker base such as potassium carbonate must be used—hydroxide or amines are no good here because they would undergo substitution reactions with the halide.



Choice of electrophile for alkylation

Enolate alkylations are S_N2 reactions (polar solvents, good charged nucleophile) so the electrophile needs to be S_N2-reactive if the alkylation is to succeed: primary and benzylic alkyl halides are among the best alkylating agents. More branched halides tend to prefer unwanted E2 elimination reactions because the anions themselves are basic. As a result, tertiary halides are useless for enolate alkylation. We shall see a way round this problem later in the chapter.

➔ Factors governing substitution reactions were covered in detail in Chapter 15, and elimination reactions were the subject of Chapter 17.

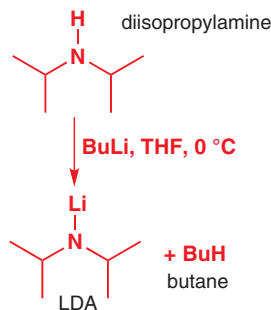
methyl	allyl	benzyl	primary alkyls	secondary alkyls	tertiary alkyls
$\text{H}_3\text{C}-\text{X}$	$\text{CH}_2=\text{CH}-\text{CH}_2-\text{X}$	$\text{C}_6\text{H}_5-\text{CH}_2-\text{X}$	$\text{R}-\text{CH}_2-\text{X}$	$\begin{matrix} \text{R}^2 \\ \\ \text{C}-\text{X} \\ \\ \text{R}^1 \end{matrix}$	$\begin{matrix} \text{R}^2 \\ \\ \text{C}-\text{X} \\ \\ \text{R}^1 \end{matrix}$
alkylate very well			alkylate well	alkylate slowly	do not alkylate

Lithium enolates of carbonyl compounds

The problem of self-condensation of carbonyl compounds (that is, enolate reacting with unenolized carbonyl) under basic conditions does not exist if there is absolutely no unenolized carbonyl compound present. One way to achieve this is to use a base sufficiently strong

➔ LDA was described on p. 465.

a reminder: how to make LDA

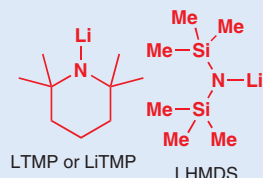


■ Enolates are a type of alkene, and may have two possible geometries. The importance of enolate geometry is discussed in Chapter 33 and will not concern us here. More important is the question of *regioselectivity* when unsymmetrical ketones are deprotonated. We shall discuss this aspect later in the chapter.

Interactive mechanism for lithium enolate formation

Variants of LDA

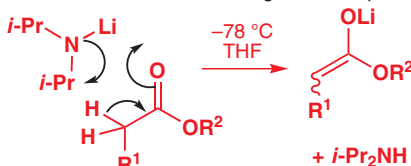
LDA came into general use in the 1970s, and you may meet variants such as those derived from butyllithium and 2,2,6,6-tetramethylpiperidine (lithium tetramethylpiperidide, LTMP) or hexamethyldisilazane (lithium hexamethyldisilazide, LHMDs), which are even more hindered and even less nucleophilic.



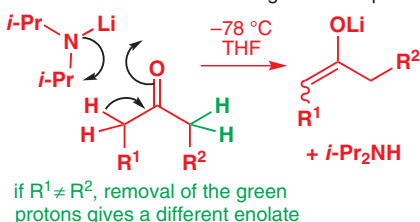
(pK_a at least 3 or 4 units higher than pK_a of the carbonyl compound) to ensure that all of the starting carbonyl is converted into the corresponding enolate. This will work only if the resulting enolate is sufficiently stable to survive until the alkylation is complete. As you saw in Chapter 20, lithium enolates are stable, and are among the best enolate equivalents for use in alkylation reactions.

The best base for making lithium enolates is usually LDA, made from diisopropylamine ($i\text{-Pr}_2\text{NH}$) and BuLi. LDA will deprotonate virtually all ketones and esters that have an acidic proton to form the corresponding lithium enolates rapidly, completely, and irreversibly even at the low temperatures (about -78°C) required for some of these reactive species to survive. Deprotonation occurs through a cyclic mechanism, which is illustrated below for ketones and esters. The basic nitrogen anion removes the proton as the lithium is delivered to the forming oxyanion.

deprotonation of an ester lithium enolate: two geometries possible



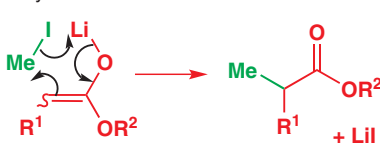
deprotonation of a ketone lithium enolate: two geometries possible



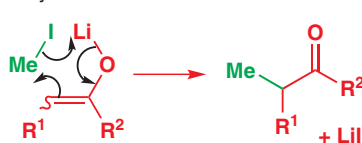
Alkylations of lithium enolates

The reaction of these lithium enolates with alkyl halides is one of the most important C–C bond-forming reactions in chemistry. Alkylation of lithium enolates works with both acyclic and cyclic ketones as well as with acyclic and cyclic esters (lactones). The general mechanism is shown below.

alkylation of an ester enolate



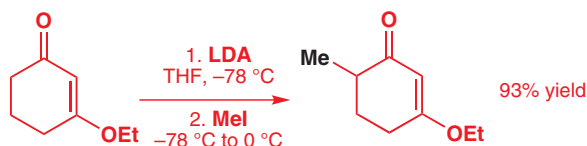
alkylation of a ketone enolate



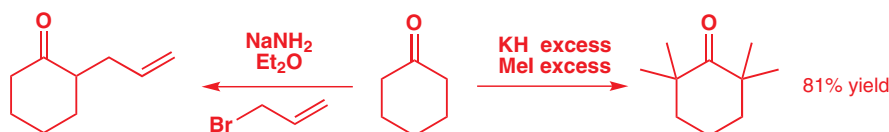
Typical experimental conditions for reactions of kinetic enolates involve formation of the enolate at very low temperature (-78°C) in THF. The strong base LDA is used to avoid self-condensation of the carbonyl compound but, while the enolate is forming, there is always a chance that self-condensation will occur. The lower the temperature, the slower the self-condensation reaction, and the fewer by-products there are. Once enolate formation is complete, the electrophile is added (still at -78°C : the lithium enolates may not be stable at higher temperatures). The reaction mixture is then usually allowed to warm up to room temperature to speed up the rate of the S_N2 alkylation.

Alkylation of ketones

Precisely this sequence was used to methylate the ketone below, with LDA acting as base followed by methyl iodide as electrophile.

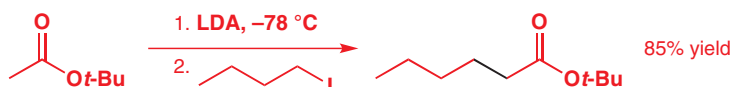


Their stability at low temperature means that lithium enolates are usually preferred, but sodium and potassium enolates can also be formed by abstraction of a proton by strong bases. The increased separation of the metal cation from the enolate anion with the larger alkali metals leads to more reactive but less stable enolates. Typical very strong Na and K bases include the hydrides (NaH, KH) or amide anions derived from ammonia (NaNH₂, KNH₂) or hexamethyldisilazane (NaHMDS, KHMDS). The instability of the enolates means that they are usually made and reacted in a single step, so the base and electrophile need to be compatible. Here are two examples of cyclohexanone alkylation: the high reactivity of the potassium enolate is demonstrated by the efficient tetramethylation with excess potassium hydride and methyl iodide.



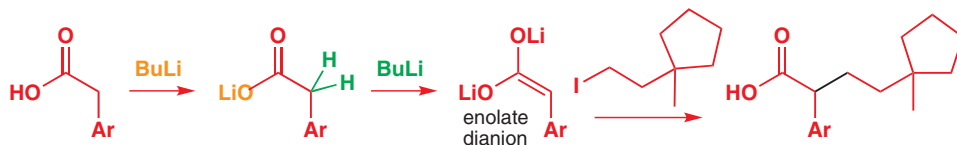
Alkylation of esters

In Chapter 26 you will meet the reaction of an ester with its own enolate: the Claisen condensation. This reaction can be an irritating side reaction in the chemistry of lithium ester enolates when alkylation is desired, and again it can be avoided only if the ester is converted entirely to its enolate under conditions where the Claisen condensation is slow. A good way of stopping this happening is to add the ester to the solution of LDA (and not the LDA to the ester) so that there is never excess ester for the enolate to react with. Another successful tactic is to make the group R as large as possible to discourage attack at the carbonyl group. Tertiary butyl esters are particularly useful in this regard because they are readily made, *t*-butyl is extremely bulky, and yet they can still be hydrolysed in aqueous acid under mild conditions by the method discussed on p. 556. In this example, deprotonation of *t*-butyl acetate gives a lithium enolate that reacts with butyl iodide as the reaction mixture is warmed to room temperature.



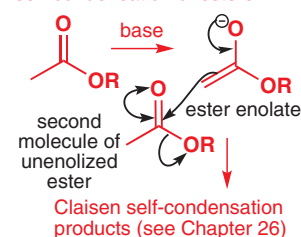
Alkylation of carboxylic acids

The lithium enolates of carboxylic acids can be formed if two equivalents of base are used: one to make the carboxylate anion and one to make the enolate. It is not necessary to use a strong base to remove the first proton but, since the second deprotonation requires a strong base such as LDA, it is often convenient to use two equivalents of LDA to form the dianion. With carboxylic acids, even BuLi can be used on occasion because the intermediate lithium carboxylate is much less electrophilic than an aldehyde or a ketone.



The next alkylation of an acid enolate is of a carbamate-protected amino acid, glycine. As you saw in Chapter 23, carbamate protecting groups are stable to basic reaction conditions. Three acidic protons are removed by LDA, but alkylation takes place only at carbon—the site of the last proton to be removed. Alkylation gets rid of one of the negative charges, so that, if the molecule gets a choice, it alkylates to get rid of the least stable anion, keeping the two more

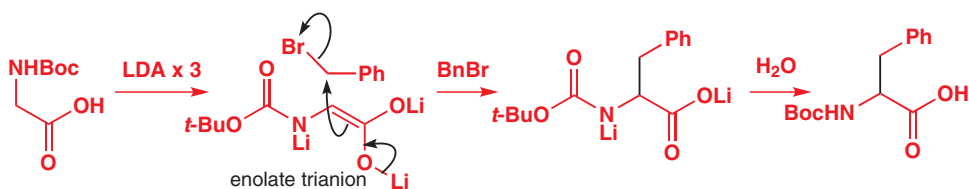
aim to avoid the Claisen self-condensation of esters



■ Why doesn't BuLi add to the carboxylate, as you saw in Chapter 10, to form the ketone? Presumably in this case the aromatic ring helps acidify the benzylic protons to tip the balance towards deprotonation. Even with carboxylic acids, LDA would normally be the first base you would try.

➡ You saw this sort of reactivity with dianions in Chapter 23: the last anion to form will be the most reactive.

stabilized charges. A good alternative to using the dianion is to alkylate the ester or nitrile and then hydrolyse to the acid.



● Alkylation of ketones, esters, and carboxylic acids is best carried out using the lithium enolates.

Why do enolates alkylate on carbon?

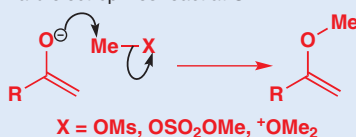
Enolates have two nucleophilic sites: the carbon and the oxygen atoms. On p. 453 we showed that:

- carbon has the greater coefficient in the HOMO, and is the softer nucleophilic site
- oxygen carries the greater total charge and is the harder nucleophilic site. In Chapter 20 you saw that hard electrophiles prefer to react at oxygen—that is why it is possible to make silyl enol ethers, for example. Some carbon electrophiles with very good leaving groups also tend to react on oxygen, but soft electrophiles such as alkyl halides react at carbon, and you will see only this type of electrophile in this chapter.

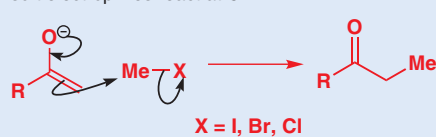
In general:

- hard electrophiles, particularly alkyl sulfates and sulfonates (mesylates, tosylates), tend to react at oxygen
- soft electrophiles, particularly alkyl halides ($I > Br > Cl$), react at carbon
- polar aprotic solvents (DMSO, DMF) promote *O*-alkylation by separating the enolate anions from each other and the counterion (making the bond more polar and increasing the charge at O) while ethereal solvents (THF, DME) promote *C*-alkylation
- larger alkali metals ($Cs > K > Na > Li$) give more separated ion pairs (more polar bonds), which are harder and react more at oxygen.

hard electrophiles react at O



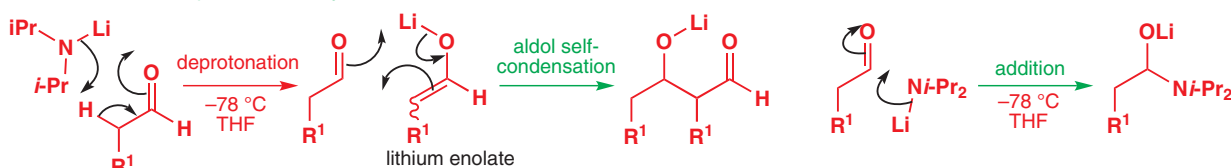
soft electrophiles react at C



Alkylation of aldehydes: avoid LDA

Aldehydes are so electrophilic that, even with LDA at -78°C , the rate at which the deprotonation takes place is not fast enough to outpace reactions between the forming lithium enolate and still-to-be-deprotonated aldehyde remaining in the mixture. Direct addition of the base to the carbonyl group of electrophilic aldehydes can also pose a problem.

reactions which compete with aldehyde enolate formation



● Avoid using lithium enolates of aldehydes.

Using specific enol equivalents to alkylate aldehydes and ketones

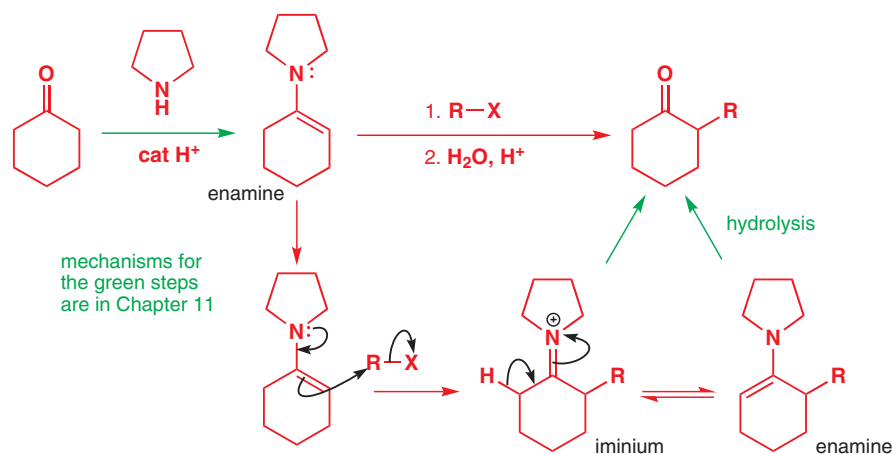
These side reactions mean that aldehyde enolates are not generally useful reactive intermediates. Instead, there are a number of aldehyde enol and enolate equivalents in which the aldehyde is present only in masked form during the enolization and alkylation step. The three most important of these specific enol equivalents are:

- enamines
- silyl enol ethers
- aza-enolates derived from imines.

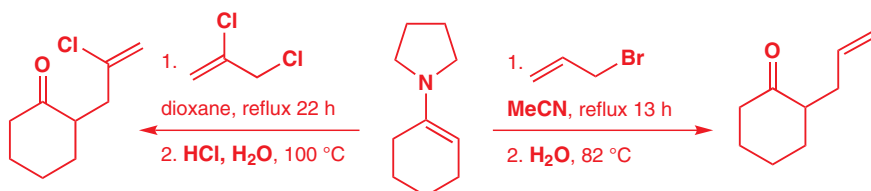
You met these enolate equivalents briefly in Chapter 20, and we shall discuss how to use them to alkylate aldehydes shortly. All three types of specific enol equivalent are useful not just with aldehydes, but with ketones as well, and we shall introduce each class with examples for both types of carbonyl compound.

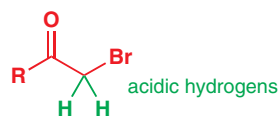
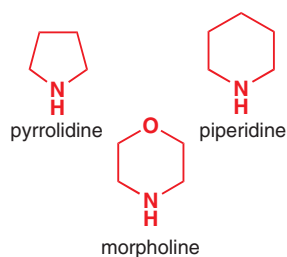
Enamines are alkylated by reactive electrophiles

Enamines are formed when aldehydes or ketones react with secondary amines. The mechanism is given in Chapter 11. The mechanism below shows how they react with alkylating agents to form new carbon–carbon bonds: the enamine here is the one derived from cyclohexanone and pyrrolidine. The product is at first not a carbonyl compound: it's an iminium ion or an enamine (depending on whether an appropriate proton can be lost). But a mild acidic hydrolysis converts the iminium ion or enamine into the corresponding alkylated carbonyl compound.



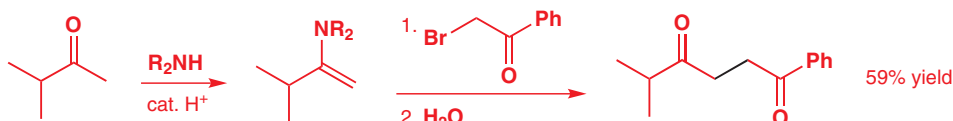
The overall process, from carbonyl compound to carbonyl compound, amounts to an enolate alkylation, but no strong base or enolates are involved so there is no danger of self-condensation. The example below shows two specific examples of cyclohexanone alkylation using an enamine. Note the relatively high temperatures and long reaction times: enamines are among the most reactive of neutral nucleophiles, but they are still a lot less nucleophilic than enolates.





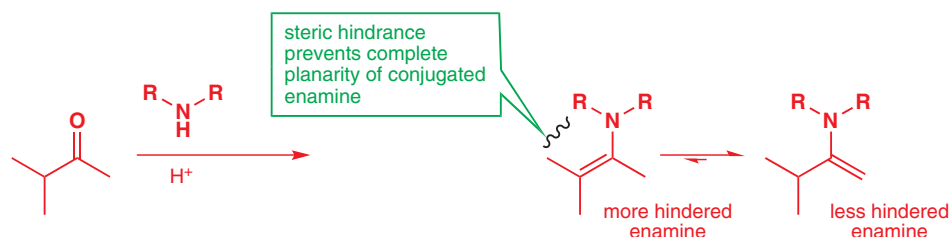
The choice of the secondary amine for formation of the enamine is not completely arbitrary even though it does not end up in the final alkylated product. Simple dialkyl amines can be used but cyclic amines such as pyrrolidine, piperidine, and morpholine are popular choices as the ring structure makes both the starting amine and the enamine more nucleophilic (the alkyl groups are 'tied back' and can't get in the way). The higher boiling points of these amines allow the enamine to be formed by heating.

α -Bromo carbonyl compounds are excellent electrophiles for S_N2 reactions because of the rate-enhancing effect of the carbonyl group (Chapter 15). The protons between the halogen and the carbonyl are significantly more acidic than those adjacent to just a carbonyl group and there can be a serious risk of an enolate nucleophile acting as a base. Enamines are only very weakly basic, but react well as nucleophiles with α -bromo carbonyl compounds, and so are a good choice.

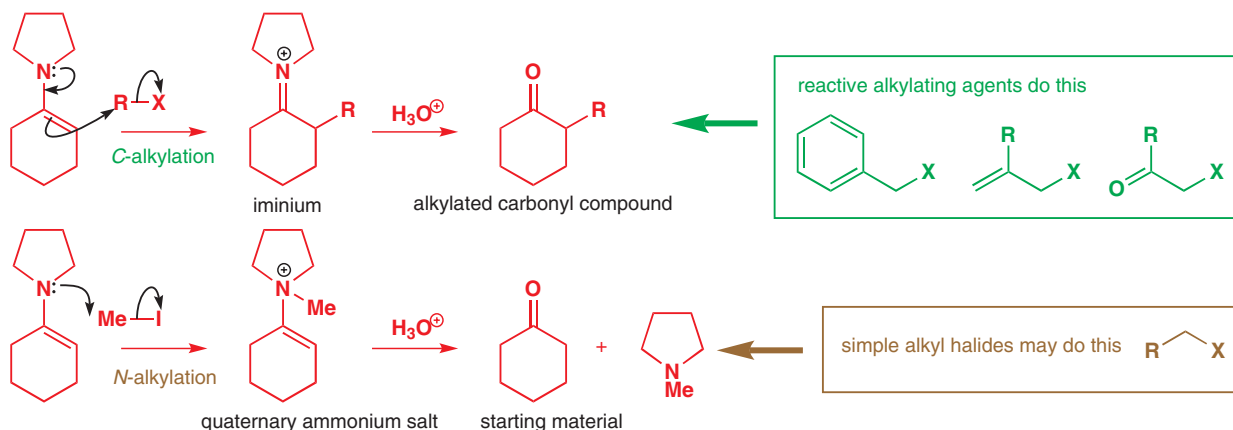


■ Note how the preference for the less substituted enamine is opposite to the preference for a more substituted enol: see p. 465.

The starting ketone here is unsymmetrical, so two enamines are possible. However, the formation of solely the *less* substituted enamine is typical. The outcome may be explained as the result of thermodynamic control: enamine formation is reversible so the less hindered enamine predominates. For the more substituted enamine, steric hindrance forces the enamine to lose planarity, and destabilizes it. The less substituted enamine, on the other hand, is rather more stable.



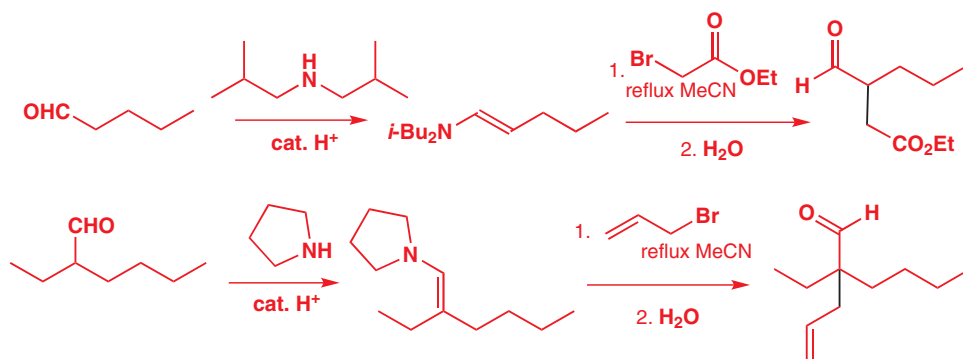
There is, however, a major problem with enamines: reaction at nitrogen. Less reactive alkylating agents—simple alkyl halides such as methyl iodide, for example—react to a significant degree at N rather than at C. The product is a quaternary ammonium salt, which hydrolyses back to the starting material and leads to low yields.



● **Enamines work best with reactive alkylating agents:**

- allylic halides
- benzyl halides
- α -halo carbonyl compounds.

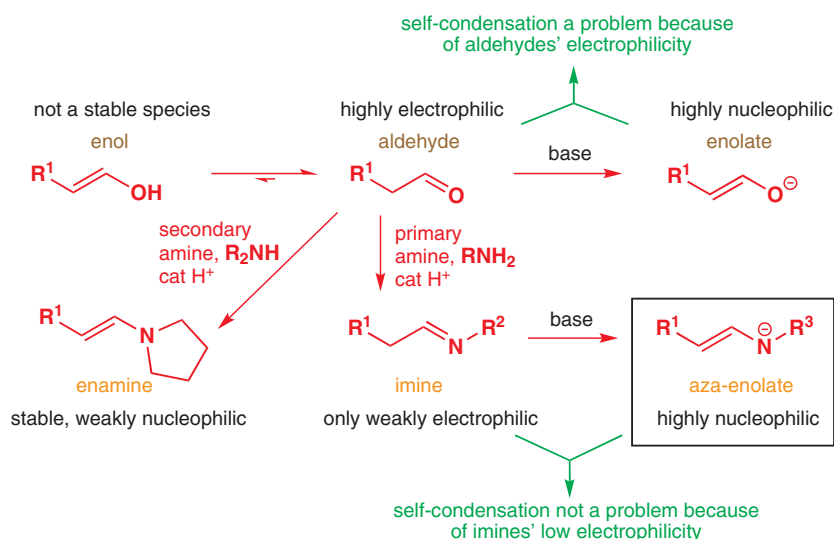
That said, enamines are a good solution to the aldehyde enolate problem. Aldehydes form enamines very easily (one of the advantages of the electrophilic aldehyde) and these are immune to attack by nucleophiles—including, most importantly, the enamines themselves. Below are two examples of aldehyde alkylation using the enamine method. Both again use highly S_N2 -reactive electrophiles, and this is the main limitation of enamines.



Aza-enolates react with a wider range of S_N2 -reactive electrophiles

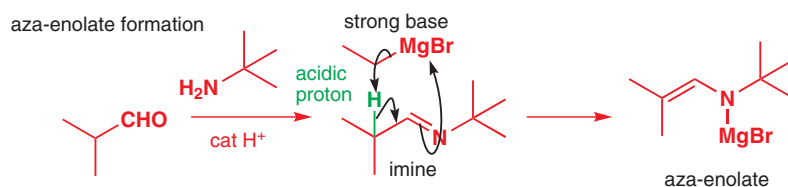
Enamines are the nitrogen analogues of enols and provide one solution to the aldehyde enolate problem when the electrophile is reactive. Imines are the corresponding nitrogen analogues of aldehydes and ketones: a little lateral thinking should therefore lead you to expect some useful reactivity from the nitrogen equivalents of enolates, known as aza-enolates. Aza-enolates are formed when imines are treated with LDA or other strong bases.

In basic or neutral solution, imines are less electrophilic than aldehydes: they react with organolithiums, but not with many weaker nucleophiles (they are more electrophilic in acid when they are protonated). So, as the aza-enolate forms, there is no danger at all of self-condensation.

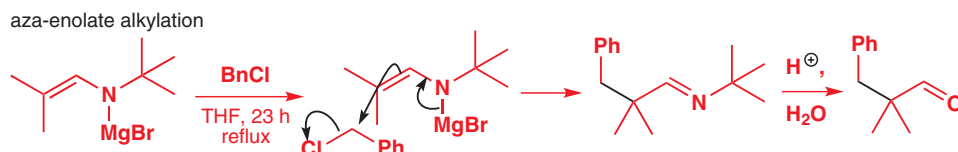


■ Note that aza-enolates are formed from imines, which can be made only from *primary* amines. Enamines are made from aldehydes or ketones with *secondary* amines.

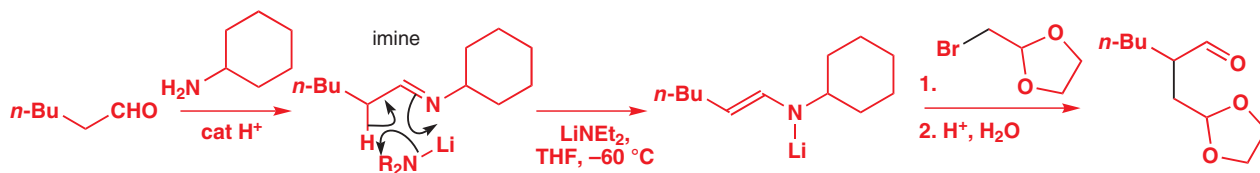
The overall sequence involves formation of the imine from the aldehyde that is to be alkylated—usually with a bulky primary amine such as *t*-butyl- or cyclohexylamine to discourage even further nucleophilic attack at the imine carbon. The imine is not usually isolated, but is deprotonated directly with LDA or a Grignard reagent (these do not add to imines, but they will deprotonate them to give magnesium aza-enolates).



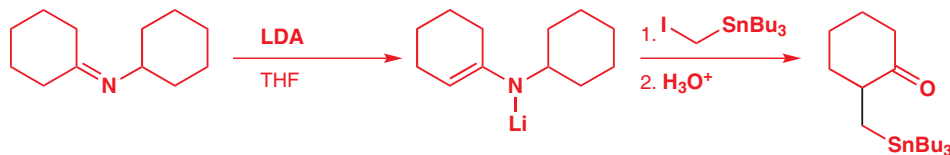
The resulting aza-enolate reacts like a ketone enolate with S_N2 -reactive alkylating agents—here, benzyl chloride—to form the new carbon–carbon bond and to re-form the imine. The alkylated imine is usually hydrolysed by the mild acidic work-up to give the alkylated aldehydes.



In the next example, a lithium base (lithium diethylamide) is used to form the aza-enolate. The ease of imine cleavage in acid is demonstrated by the selective hydrolysis to the aldehyde without any effect on the acetal introduced by the alkylation step. The product is a mono-protected dialdehyde, which is difficult to prepare by other methods.



Aza-enolate alkylation is so successful that it has been extended from aldehydes, where it is essential, to ketones, where it can be a useful option. Cyclohexanones are among the most electrophilic simple ketones and can suffer from undesirable side reactions. The imine from cyclohexanone and cyclohexylamine can be deprotonated with LDA to give a lithium aza-enolate. In this example, iodomethylstannane was the alkylating agent, giving the tin-containing ketone after hydrolysis.

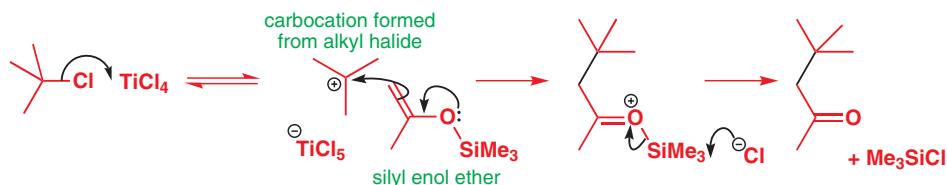


• Aldehyde alkylation

Aza-enolates are the best general solution for alkylating aldehydes with most electrophiles. With very S_N2 -reactive alkylating agents, enamines can be used, and with very S_N1 -reactive alkylating agents, silyl enol ethers must be used.

Silyl enol ethers are alkylated by S_N1 -reactive electrophiles in the presence of Lewis acid

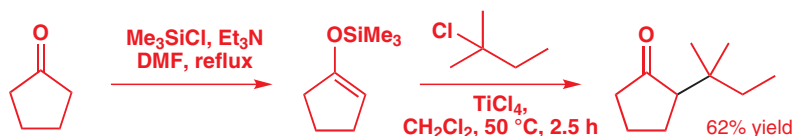
While the greater nucleophilicity of azaenolates means that they will react with a wider range of electrophiles, their basicity, like that of lithium enolates, means that they will not react with S_N1 -reactive electrophiles like tertiary alkyl halides. The solution to this problem is to use silyl enol ethers, which are less reactive and so require a more potent electrophile to initiate reaction. Carbocations will do, and they can be generated *in situ* by abstraction of a halide or other leaving group from a saturated carbon atom.



➔ You met silyl enol ethers in Chapter 20, p. 466.

■ TiCl₄ is acting as a Lewis acid here (see p. 180 for more on Lewis acids), accepting a pair of electrons from the Cl atom. You saw the quantitative formation of carbocations by a related method in Chapter 15.

The best alkylating agents for silyl enol ethers are tertiary alkyl halides: they form stable carbocations in the presence of Lewis acids such as TiCl₄ or SnCl₄. Most fortunately, this is just the type of compound that is unsuitable for reaction with lithium enolates or enamines, as elimination results rather than alkylation: a nice piece of complementary selectivity. Below is an example: the alkylation of cyclopentanone with 2-chloro-2-methylbutane. The ketone was converted to the trimethylsilyl enol ether with triethylamine and trimethylsilylchloride: we discussed this step on p. 466 (Chapter 20). Titanium tetrachloride in dry dichloromethane promotes the alkylation step.



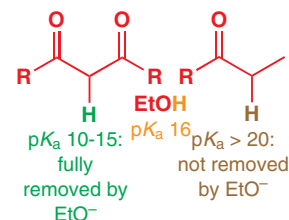
Summary: specific enol equivalents for aldehydes and ketones:

- Lithium enolates can be used with S_N2 -reactive electrophiles, but cannot be made from aldehydes.
- Aza-enolates of aldehydes or ketones can be used with the same S_N2 -reactive electrophiles, but can be made from aldehydes.
- Enamines of aldehydes or ketones can be used with allylic, benzylic, or α -halocarbonyl compounds.
- Silyl enol ethers of aldehydes or ketones can be used with S_N1 -reactive electrophiles such as allylic, benzylic, or tertiary alkyl halides.

Alkylation of β -dicarbonyl compounds

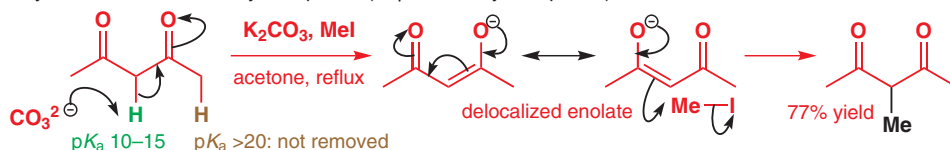
The presence of two, or even three, electron-withdrawing groups on a single carbon atom makes the remaining proton(s) sufficiently acidic (pK_a 10–15) that even mild bases can lead to complete enolate formation. Bases of the strength of alkoxides (pK_a of ROH = ca. 16) cannot deprotonate simple carbonyl compounds (pK_a 20–25) completely, but readily generate anions stabilized by more than one electron-withdrawing group. The most important enolates of this type are those of 1,3-dicarbonyl (or β -dicarbonyl) compounds.

The resulting anions are alkylated very efficiently. This diketone is enolized even by potassium carbonate, and reacts with methyl iodide in good yield. Carbonate is such a bad nucleophile that the base and the electrophile can be added in a single step.



Enolates stabilized by two of the following electron-withdrawing groups may be formed with alkoxides: COR, CO₂R, CN, CONR₂, SO₂R, (RO)₂P=O.

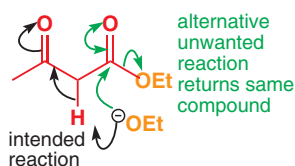
alkylation of a 1,3-dicarbonyl compound (or β -dicarbonyl compound)



Among the β -dicarbonyls two compounds stand out in importance—diethyl (or dimethyl) malonate and ethyl acetoacetate. You should make sure you remember their structures and trivial names.

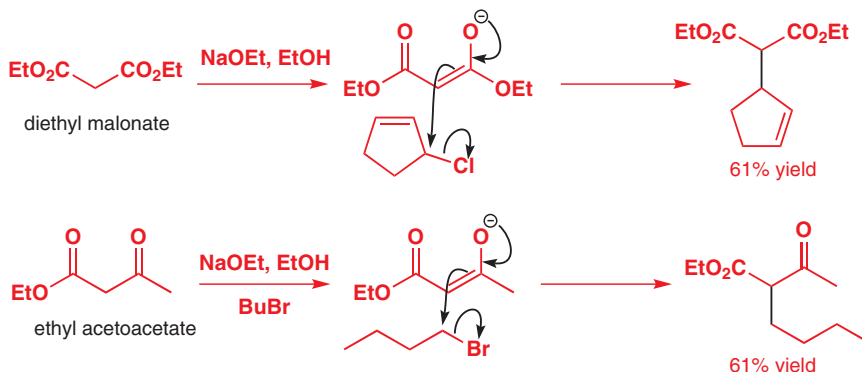


➡ You met the stable enols of related compounds in Chapter 20.



With these two esters, the choice of base is important: nucleophilic addition can occur at the ester carbonyl, which could lead to transesterification (with alkoxides), hydrolysis (with hydroxide), or amide formation (with amide anions). The best choice is usually an alkoxide identical with the alkoxide component of the ester (that is, *ethoxide* for diethyl malonate, *methoxide* for dimethyl malonate). Alkoxides are basic enough to deprotonate between two carbonyl groups but, should substitution occur at $\text{C}=\text{O}$, there is no overall reaction.

In the first example below the electrophile is the allylic cyclopentenyl chloride, and the base is ethoxide in ethanol—most conveniently made by adding one equivalent of sodium metal to dry ethanol. The same base is used in the second alkylation, of ethyl acetoacetate with butyl bromide.



Various electron-withdrawing groups can be used in almost any combination with good results. In this example an ester and a nitrile cooperate to stabilize an anion. Nitriles are not quite as anion-stabilizing as carbonyl groups so this enolate requires a stronger base (sodium hydride) in an aprotic solvent (DMF) for success. The primary alkyl tosylate serves as the electrophile.

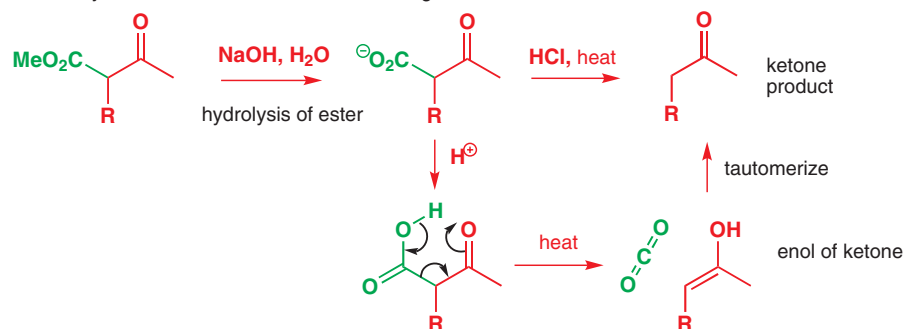
➡ If you need a reminder about the tosylate leaving group, turn back to p. 349.



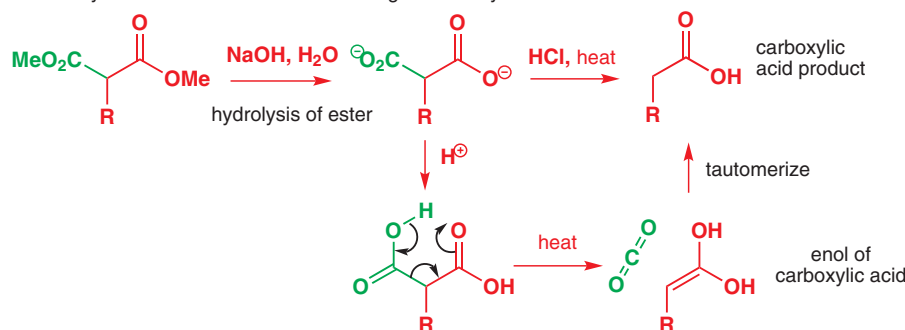
These doubly stabilized anions are alkylated so well that it is common to carry out an alkylation between two carbonyl groups, only to remove one of them at a later stage. This is made possible by the fact that carboxylic acids with a β -carbonyl group decarboxylate (lose carbon dioxide) on heating. The mechanism below shows how. After alkylation of the dicarbonyl compound the unwanted ester is first hydrolysed in base. Acidification and heating lead to

decarboxylation via a six-membered cyclic transition state in which the acid proton is transferred to the carbonyl group as the key bond breaks, liberating a molecule of carbon dioxide. The initial product is the enol form of a carbonyl compound that rapidly tautomerizes to the more stable keto form—now with only one carbonyl group. Using this technique, β -keto-esters give ketones while malonate esters give simple carboxylic acids (both ester groups hydrolyse but only one can be lost by decarboxylation). Decarboxylation can occur only with a second carbonyl group appropriately placed β to the acid, because the decarboxylated product must be formed as an enol.

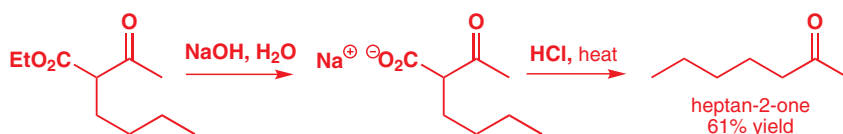
decarboxylation of acetoacetate derivatives to give ketones



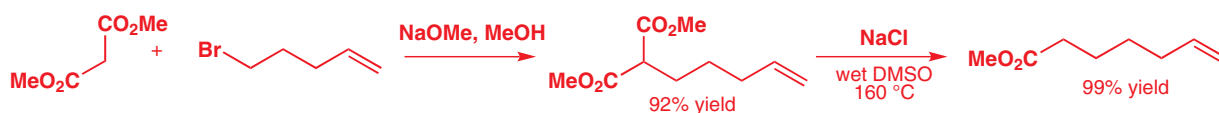
decarboxylation of malonate derivatives to give carboxylic acids



The alkylation of ethyl acetoacetate with butyl bromide on p. 596 was done with the expressed intention of decarboxylating the product to give hexan-2-one. These are the conditions for this decarboxylation: the heating step drives off the CO_2 by increasing the gearing on the entropy term ($\Delta S^\ddagger$) of the activation energy (two molecules are made from one).



Esters are much easier to work with than carboxylic acids, and a useful alternative procedure removes one ester group without having to hydrolyse the other. The malonate ester is heated in a polar aprotic solvent—usually DMSO—in the presence of sodium chloride and a little water. No acid or base is required and, apart from the high temperature, the conditions are fairly mild. The scheme below shows a dimethyl malonate alkylation (note that NaOMe is used with the dimethyl ester) and removal of the methyl ester.

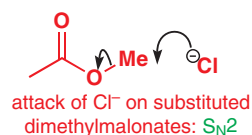
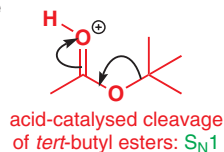
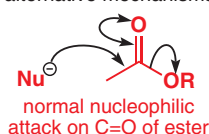


➡ We discussed the role of temperature in driving reactions in Chapter 12.

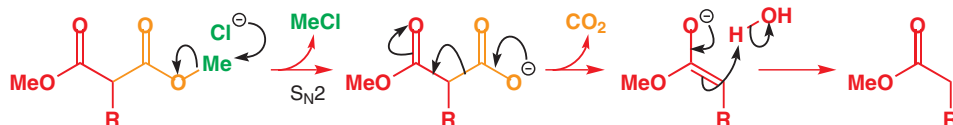
■ *tert*-Butyl esters also typically hydrolyse by cleavage of the O–alkyl bond, as we showed you on p. 556. With a *t*-butyl group the mechanism is of course S_N1 .

The mechanism of decarboxylation is a rather unusual type of ester cleavage reaction. The bond that breaks is not the $\text{MeO}-\text{CO}$ bond but instead the O–alkyl bond: the reaction is an S_N2 substitution of carboxylate by Cl^- .

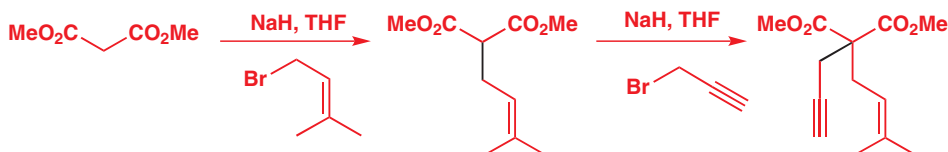
alternative mechanisms for ester cleavage



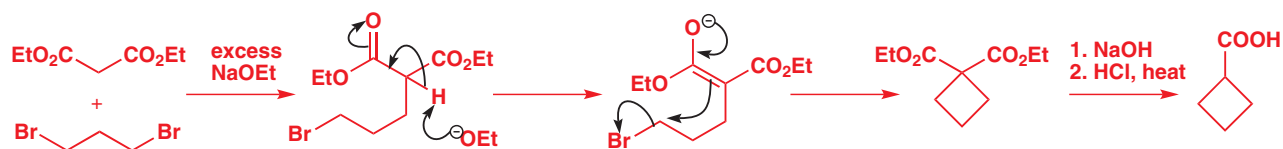
Chloride is a poor nucleophile, but it is more reactive in DMSO, by which it cannot be solvated. And, as soon as the carboxylate is displaced, the high temperature encourages (entropy again) irreversible decarboxylation. The other by-product, MeCl , is also lost as a gas. The ‘decarboxylation’ (in fact, removal of a CO_2Me group, not CO_2) is known as the **Krapcho decarboxylation**. Because of the S_N2 step, it works best with *methyl* malonate esters.



We have only looked at single alkylations of dicarbonyl compounds, but there are two acidic protons between the carbonyl groups and a second alkylation is usually possible. Excess of base and alkyl halide gives two alkylations in one step. More usefully, it is possible to introduce two different alkyl groups by using just one equivalent of base and alkyl halide in the first step.

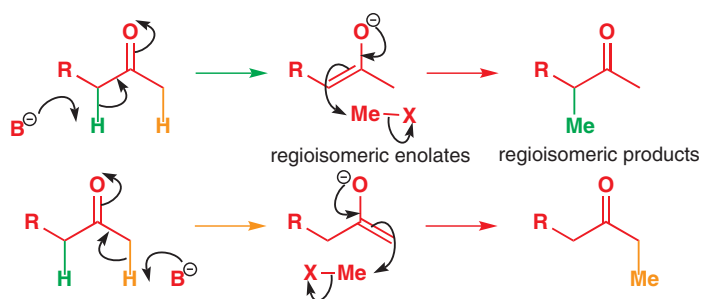


With a dihaloalkane, rings can be formed by two sequential alkylation reactions: this is an important way of making cycloalkanecarboxylic acids. Even the usually more difficult (see Chapter 31) four-membered rings can be made in this way.



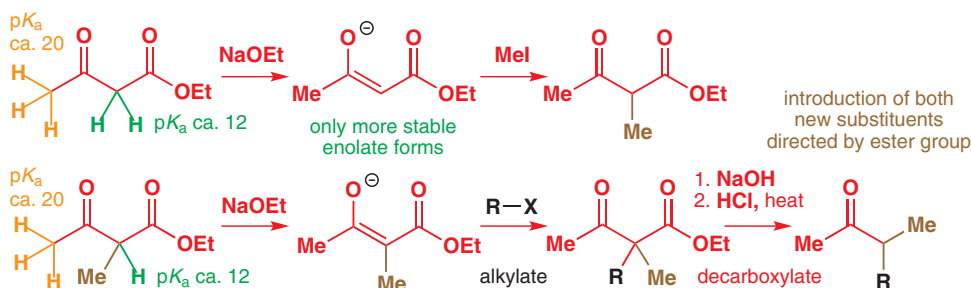
Ketone alkylation poses a problem in regioselectivity

Ketones are unique because they can have enolizable protons on both sides of the carbonyl group. Unless the ketone is symmetrical, or unless one side of the ketone happens to have no enolizable protons, two regioisomers of the enolate are possible and alkylation can occur on either side to give regioisomeric products. We need to be able to control which enolate is formed if ketone alkylations are to be useful.

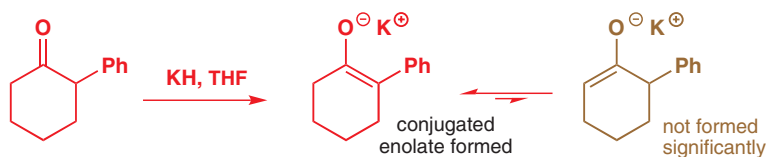


Thermodynamically controlled enolate formation

Selective enolate formation is straightforward if the protons on one side of the ketone are significantly more acidic than those on the other. This is what you have just seen with ethyl acetoacetate: it is a ketone, but with weak bases (pK_a of the conjugate acid < 18) it only ever enolizes on the side where the protons are acidified by the second electron-withdrawing group. If two new substituents are introduced, in the manner you have just seen, they will always both be joined to the same carbon atom. This is an example of thermodynamic control: only the more stable of the two possible enolates is formed.



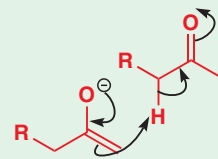
This principle can be extended to ketones whose enolates have less dramatic differences in stability. Since enols and enolates are alkenes, the more substituents they carry the more stable they are. So, in principle, even additional alkyl groups can control enolate formation under thermodynamic control. Formation of the more stable enolate requires a mechanism for equilibration between the two enolates, and this must be proton transfer. If a proton source is available—and this can even be just excess ketone—an equilibrium mixture of the two enolates will form. The composition of this equilibrium mixture depends very much on the ketone but, with 2-phenylcyclohexanone, conjugation ensures that only one enolate forms. The base is potassium hydride: it's strong, but small (and so has no difficulty removing the more hindered proton) and can be used under conditions that permit enolate equilibration.

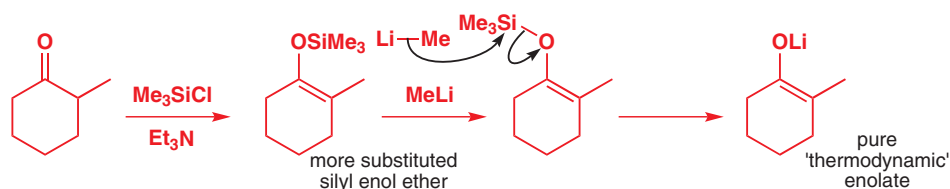


The more substituted lithium enolates can also be formed from the more substituted silyl enol ethers by substitution at silicon—a reaction you met in Chapter 20. The value of this reaction now becomes clear because the usual way of making silyl enol ethers (Me_3SiCl , Et_3N) typically produces, from unsymmetrical ketones, the more substituted of the two possible ethers. Because the silyl enol ether (unlike the enolate itself) can be purified, fully regiochemically pure enolates can be formed in this way.

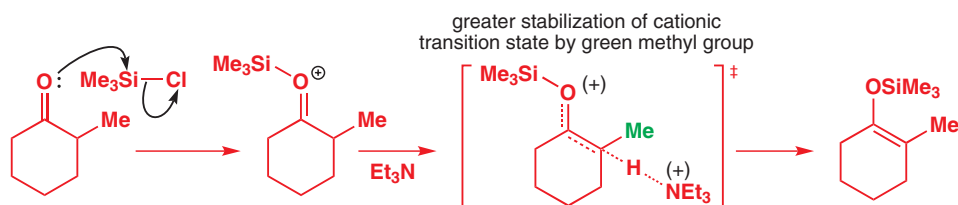
➡ The influence of substituents on the stability of alkenes was discussed on p. 394. The fact that more substituted enols are more stable was discussed in Chapter 20, p. 465.

■ The mechanism for equilibration simply involves deprotonation of a molecule of ketone (the proton source) by a molecule of enolate:

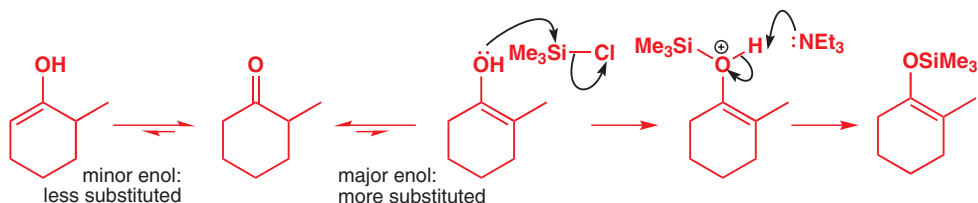




One possible explanation for the thermodynamic regioselectivity in the enol ether-forming step is related to our rationalization of the regioselectivity of bromination of ketones in acid on p. 464. Triethylamine is too weak a base (pK_a of Et_3NH^+ is about 10) to deprotonate the starting carbonyl compound (pK_a ca. 20), and the first stage of the reaction is probably an oxygen–silicon interaction. Loss of a proton now takes place through a cationic transition state, and this is stabilized rather more if the proton being lost is next to the methyl group: methyl groups stabilize partial cations just as they stabilize cations.



An alternative view is that reaction takes place through the enol: the Si–O bond is so strong that even neutral enols react with Me_3SiCl , on oxygen, of course. The predominant enol is the more substituted, leading to the more substituted silyl enol ether.



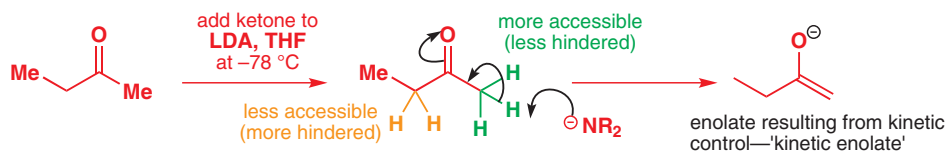
➡ Kinetic and thermodynamic control were discussed in Chapters 12, 23, and 24, pp. 264, 546, and 581.

■ To understand why less substituted C atoms have more acidic C–H bonds, think of base strengths: MeLi is a weaker base than $t\text{-BuLi}$, so the conjugate acid must be a stronger acid.

There must never be more ketone in the mixture than base, or exchange of protons between ketone and enolate will lead to equilibration. Kinetic enolate formations with LDA must be done by adding the ketone to the LDA so that there is excess LDA present throughout the reaction.

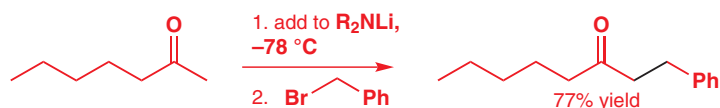
Kinetically controlled enolate formation

LDA is too hindered to attack most carbonyl $\text{C}=\text{O}$ bonds, so it attacks C–H instead. And, if there is a choice of C–H bonds, it will attack the least hindered possible. It will also prefer to attack more acidic C–H bonds, and C–H bonds on less substituted carbons are indeed more acidic. Furthermore, statistics helps, since a less substituted C atom has more protons to be removed (three versus two in this example) so, even if the rates were the same, the less substituted enolate would predominate.

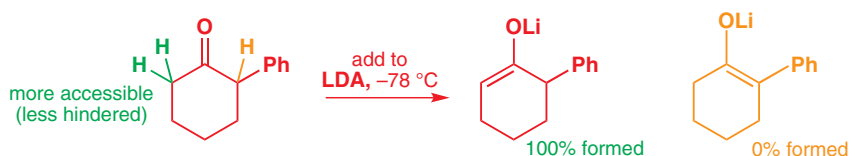


These factors multiply to ensure that the enolate that forms will be the one with the fewer substituents—provided we now prevent equilibration of the enolate to the more stable, more substituted one. This means keeping the temperature low, typically -78°C , keeping the reaction time short, and using an excess of strong base to deprotonate irreversibly and ensure that there is no remaining ketone to act as a proton source. The enolate that we then get is the one that formed faster, under kinetic control—known as the 'kinetic enolate'—and not necessarily the one that is more stable.

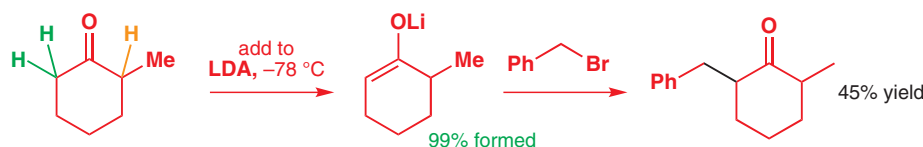
In general, this effect is sufficient to allow selective kinetic deprotonation of methyl ketones, that is, where the distinction is between Me and alkyl:



The same method works very well for 2-substituted cyclohexanones: the less substituted enolate forms. Even with 2-phenylcyclohexanone, which, as you have just seen, has a strong thermodynamic preference for the conjugated enolate, only the less substituted enolate forms.



2-Methylcyclohexanone can be regioselectively alkylated using LDA and benzyl bromide by this method.



Regioselective formation of enolates from ketones

Thermodynamic enolates are:

- more substituted
- more stable
- favoured by excess ketone, high temperature, long reaction time

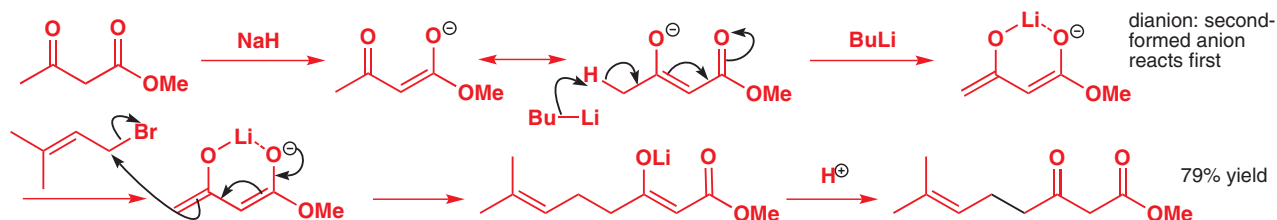
Kinetic enolates are:

- less substituted
- less stable
- favoured by strong, hindered base (e.g. LDA), low temperature, short reaction time

Dianions allow unusual regioselectivity in alkylations of methyl acetoacetate

In Chapter 23, we introduced the idea that the last-formed anion in any dianion or trianion is the most reactive. Methyl acetoacetate is usually alkylated on the central carbon atom because that is the site of the most stable enolate. But methyl acetoacetate dianion—formed by removing a second proton from the usual enolate with a very strong base (usually butyllithium)—reacts first on the less stable anion: the terminal methyl group. Protonation of the more stable enolate then leads to the product. Butyllithium can be used as a base because the anionic enolate intermediate is not electrophilic.

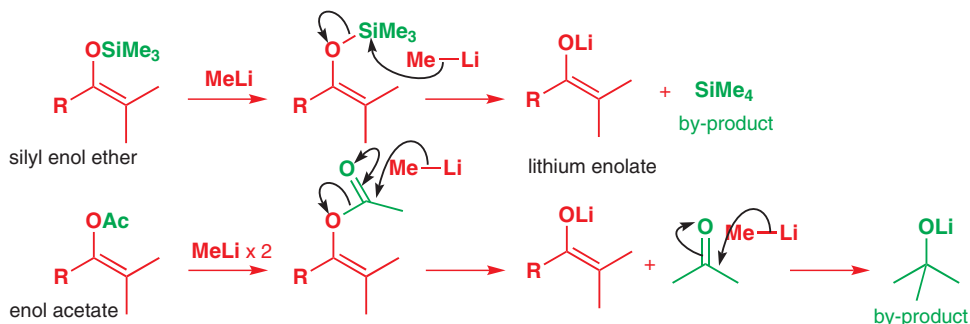
➡ Dianions were discussed on p. 547.



Enones provide a solution to regioselectivity problems

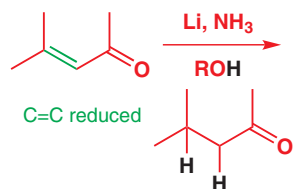
Enolates can be made regiospecifically from, for example, silyl enol ethers or enol acetates just by treating them with an alkylolithium. These are both substitution reactions in which RLi

displaces the enolate: one is $S_N2(\text{Si})$ and the other is attack at $\text{C}=\text{O}$. Provided there is no proton source, the enolate products have the same regiochemistry as their stable precursors, and single enolate regioisomers are formed.



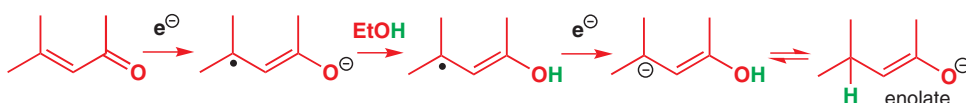
But there is a problem: forming enol ethers or enol esters will usually itself require a regioselective enolization! There are two situations in which this method is nonetheless useful: when the *more* substituted lithium enolate (which is hard to make selectively otherwise) is required, and when a silyl enol ether can be formed by a method not involving deprotonation. These methods are what we shall now consider.

Dissolving metal reduction of enones gives enolates regioselectively

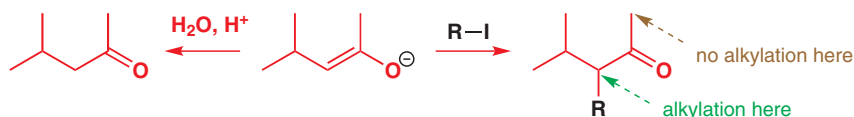


In Chapter 23 you met the Birch reduction: the use of dissolving metals (K, Na, or Li in liquid ammonia, for example) to reduce aromatic rings and alkynes. The dissolving metal reduction of enones by lithium metal in liquid ammonia is similar to these reactions—the $\text{C}=\text{C}$ bond of the enone is reduced, with the $\text{C}=\text{O}$ bond remaining untouched. An alcohol is required as a proton source and, in total, two electrons and two protons are added in a stepwise manner, giving net addition of a molecule of hydrogen to the double bond.

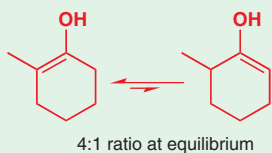
The mechanism follows that described on p. 543: transfer of an electron forms a radical anion that is protonated by the alcohol to form a radical. A second electron transfer forms an anion that can undergo tautomerization to an enolate.



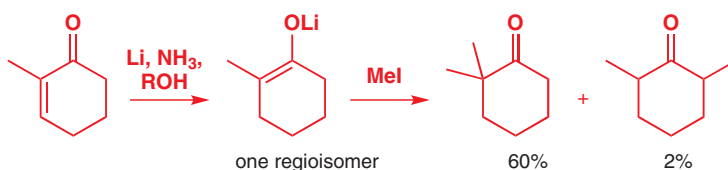
The enolate is stable to further reduction and protonation during the work-up will give a ketone. But reaction with an alkyl halide is more fruitful: because the enolate forms only where the double bond of the enone was, regioselective alkylation becomes possible.



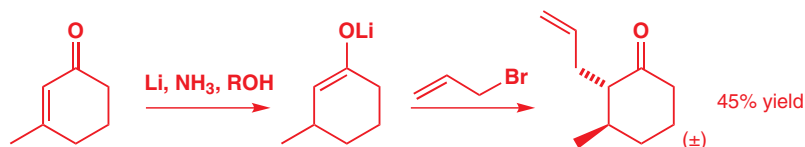
■ Thermodynamic control gives a 4:1 ratio of the two enols.



The example below leads to the regioselective methylation of methylcyclohexanone. Only 2% of the minor regioisomer is formed.



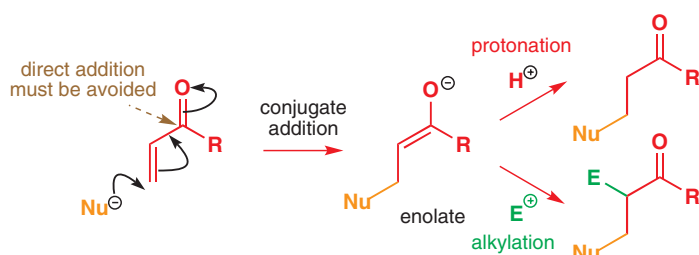
The transfer of electrons is not susceptible to steric hindrance so substituted alkenes pose no problem. In the next example, the enolate reacts with allyl bromide to give a single diastereoisomer of the product (the allyl bromide attacks from the face opposite the methyl group). Naturally, only one regioisomer is formed as well.



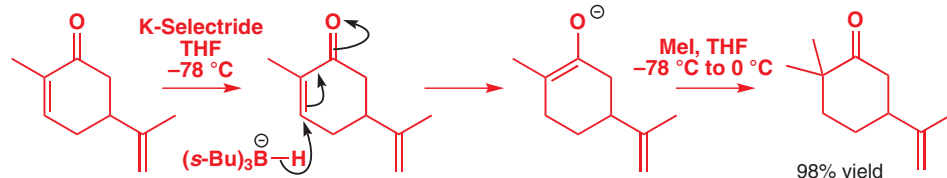
➡ We will discuss *stereo-selective* formation of single diastereoisomers in much more detail in Chapters 32 and 33.

Conjugate addition to enones gives enolates regioselectively

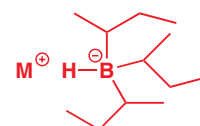
Although we did not talk in detail about them at that time, you will recall from Chapter 22 that conjugate addition to enones generates first an enolate, which is usually protonated in the work-up. But, again, more fruitful things can be done with the enolate under the right conditions.



The simplest products are formed when Nu=H, but this poses a problem of regioselectivity in the nucleophilic attack step: a nucleophilic hydride equivalent that selectively undergoes conjugate addition to the enone is required. This is usually achieved with extremely bulky hydride reagents such as lithium or potassium tri(*sec*-butyl)borohydride (often known by the trade names of L- or K-Selectride, respectively). In this example, K-Selectride reduces the enone to an enolate that is alkylated by methyl iodide to give a single regioisomer.



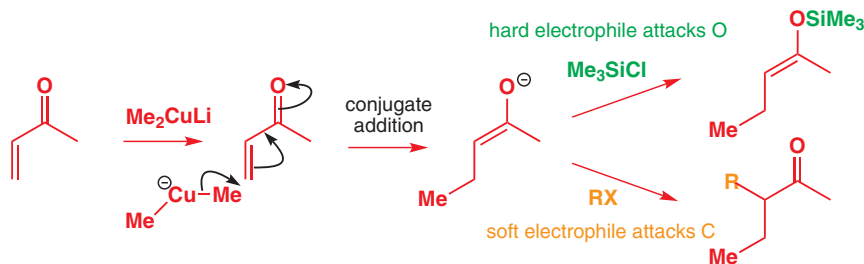
bulky reducing agents



M = Li: lithium tri-*sec*-butylborohydride (L-Selectride)
M = K: potassium tri-*sec*-butylborohydride (K-Selectride)

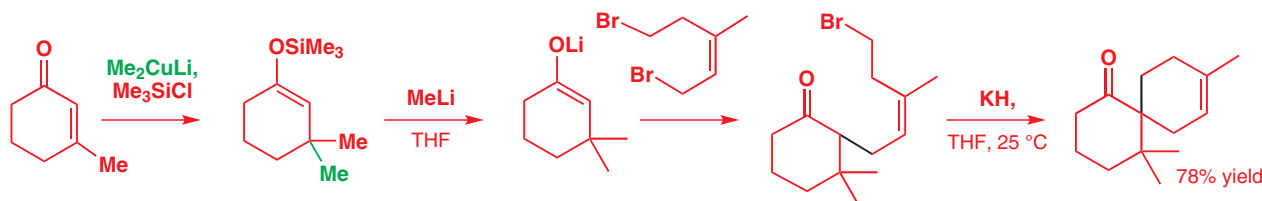
■ The reaction also illustrates the difference in reactivity between conjugated and isolated double bonds.

With organocopper reagents, conjugate addition introduces a new alkyl group and, if the resulting enolates are themselves alkylated, two new C—C bonds can be formed in a single step (a tandem reaction: one C—C bond formation rides behind another). In Chapter 22 we explained that the best organocuprate additions are those carried out in the presence of Me₃SiCl: the product of these reactions is a silyl enol ether, formed regioselectively (the 'enol' double bond is always on the side where the enone used to be).

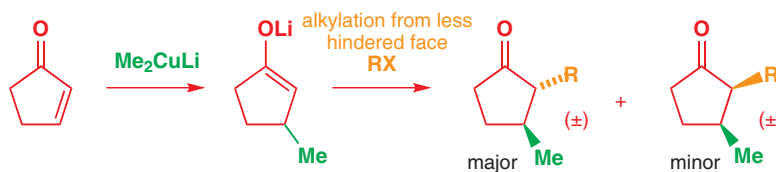


The silyl enol ethers are too unreactive for direct alkylation by an alkyl halide, but by converting them to lithium enolates all the usual alkylation chemistry becomes possible. This type of reaction forms the key step in a synthesis of the natural product α -chamigrene. Conjugate addition of Me_2CuLi gives an enolate that is trapped with trimethylsilyl chloride. Methyl lithium converts the resulting silyl enol ether into a lithium enolate by substitution at Si. The natural product has a *spiro* six-membered ring attached at the site of the enolate, and this is made by alkylating with a dibromide (you saw this done on p. 598). The first substitution is at the more reactive allylic bromide. A second enolization is needed to make the ring, but this can be done under equilibrating conditions because the required six-membered ring forms much faster than the unwanted eight-membered ring that would arise by attack on the other side of the ketone.

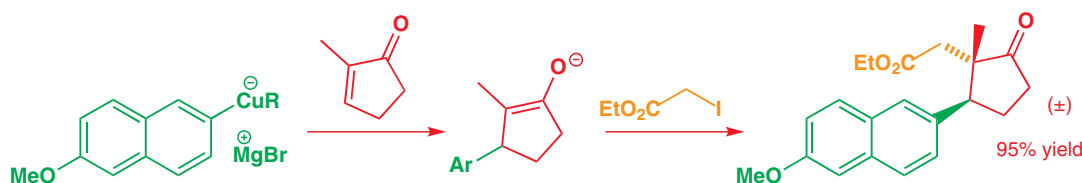
➡ The ease with which different ring sizes form is discussed in Chapter 31.



Among the most important of these tandem conjugate addition–alkylation reactions are those of cyclopentenones. With cyclopentenone itself, the *trans* diastereoisomer usually results because the alkylating agent approaches from the less hindered face of the enolate.

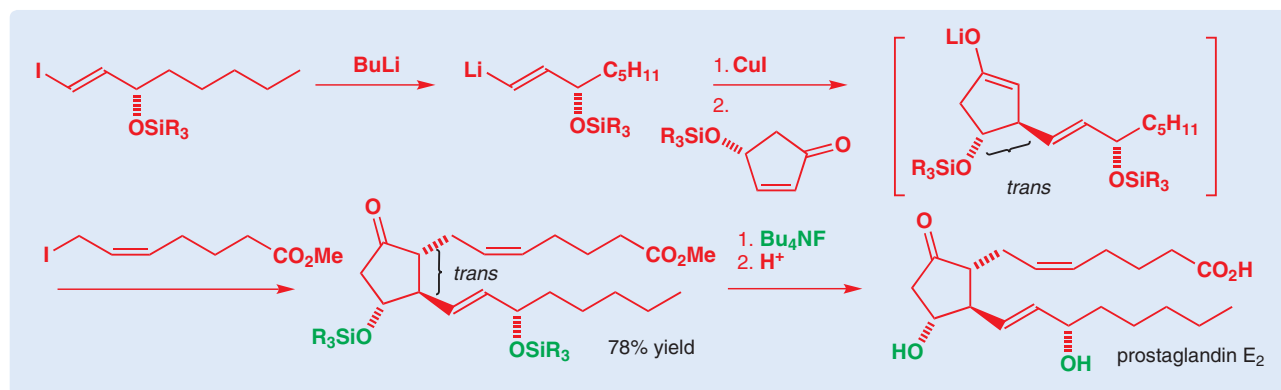


This is the sort of selectivity evident in the next example, which looks more complicated but is really just addition of an arylcopper reagent followed by alkylation (*trans* to the bulky Ar group) with an iodoester.



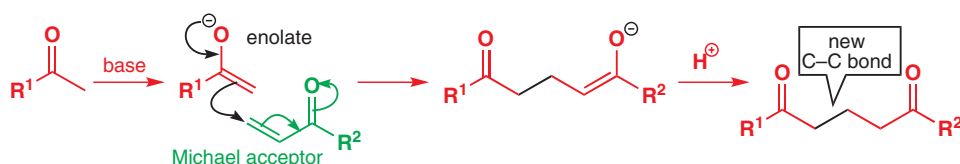
Synthesis of prostaglandin E_2

One of the most dramatic illustrations of the power of conjugate addition followed by alkylation is the short synthesis of the important biological molecule prostaglandin E_2 by Ryoji Noyori in Japan. The organocopper reagent and the alkylating agent contain all the functionality required for both side chains of the target in protected form. The required *trans* stereochemistry is assembled in the key step, which gives a 78% yield of a product requiring only removal of the silyl ether and ester protecting groups. The organometallic nucleophile was prepared from a vinyl iodide by halogen–metal exchange (Chapter 9). In the presence of copper iodide this vinyl lithium adds to the cyclopentenone in a conjugate sense to give an intermediate enolate. Because in this case the starting enone already has a stereogenic centre, this step is also stereoselective: attack on the less hindered face (opposite the silyl ether) gives the *trans* product. The resulting enolate was alkylated with the allylic iodide containing the terminal ester: once again the *trans* product was formed. It is particularly vital that enolate equilibration is avoided in this reaction to prevent the inevitable E1cB elimination of the silyloxy group that would occur from the other enolate. Deprotection of the silyl groups using TBAF (Chapter 23) gives the product.

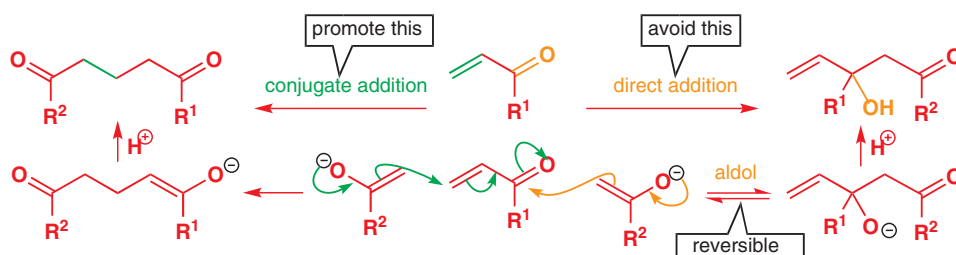


Using Michael acceptors as electrophiles

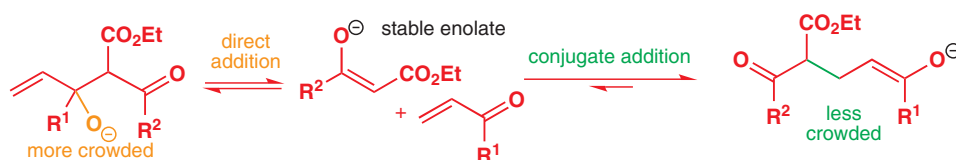
α,β -Unsaturated carbonyl compounds are, as you have just seen, an excellent source of regio-defined enolate equivalents. But they are also very effective electrophiles for reaction *with* enolates. In this last section we will consider conjugate addition reactions of enolates as an alternative way of making C–C bonds.



As with other conjugate additions, it is important in such reactions to choose conditions that prevent the nucleophile (here the enolate) attacking the C=O group directly. The same factors discussed in Chapter 22 govern the eventual outcome of the reaction. Thermodynamic control leads to conjugate addition but kinetic control leads to direct addition, so the key to successful conjugate addition is to ensure that direct addition to the carbonyl group is reversible. This enables the conjugate addition to compete and, as its product is more stable (it loses the weaker C=C π bond rather than the stronger C=O π), it eventually becomes the sole product.



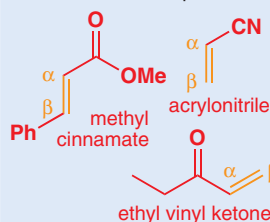
One of the most important ways of making the direct addition reversible is to use a more stabilized enolate, since expulsion of the stable anion from the direct addition product is more favourable. An additional consequence of adding a second electron-withdrawing group such as CO₂Et is that the direct addition product is more hindered (and therefore less stable) than the conjugate addition product.



The nature of the carbonyl group in the α,β -unsaturated electrophile is also important as the more electrophilic carbonyl groups give more direct addition and the less electrophilic

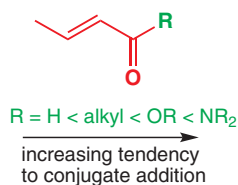
A reminder from Chapter 22: a Michael acceptor is a compound capable of undergoing conjugate addition—an α,β unsaturated carbonyl compound or nitrile for example. Many Michael acceptors are toxic and carcinogenic compounds, and must be handled with care.

some Michael acceptors...



➡ Direct attack of an enolate on a C=O group—the aldol reaction—will be the subject of the next chapter.

🔗 Interactive mechanism for conjugate addition of enolates



carbonyl groups (esters, amides) give more conjugate addition. Aldehydes and ketones can be pushed towards conjugate addition pathways by careful choice of enolate equivalent, while esters and amides are much less electrophilic at the carbonyl carbon and so are good substrates for conjugate addition.

● **Conjugate addition is thermodynamically controlled; direct addition is kinetically controlled.**

Stabilized enolates promote conjugate addition by:

- making direct addition (aldol reaction) more reversible
- making the direct addition (aldol) product more hindered.

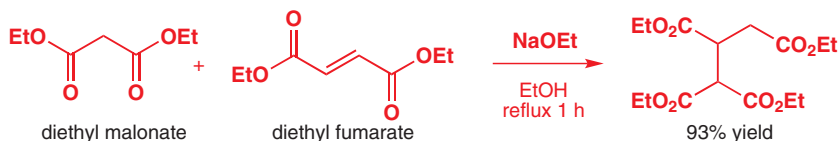
Less reactive Michael acceptors promote conjugate addition by:

- making direct addition (aldol reaction) more reversible
- making the carbonyl group less electrophilic.

1,3-Dicarbonyl compounds undergo conjugate addition

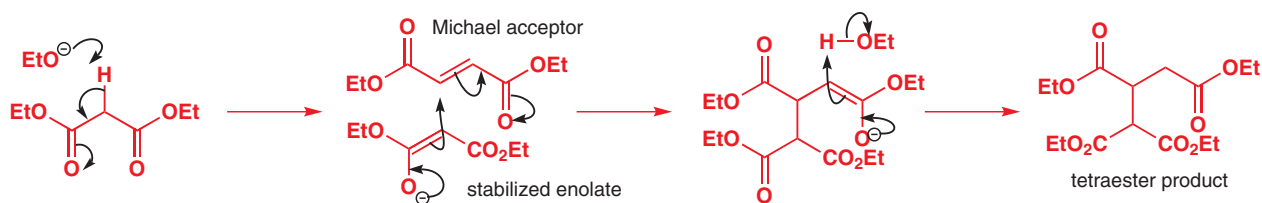
β -Diesters (malonates and substituted derivatives, see p. 595) combine three useful features in conjugate addition reactions:

- they form stable enolate anions that undergo clean conjugate addition
- if required, one of the ester groups can be removed by hydrolysis and decarboxylation
- the remaining acid or ester is ideal for conversion into other functional groups.

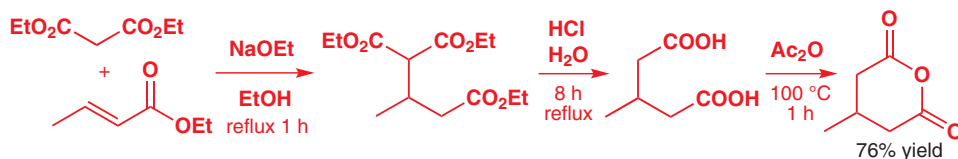


➡ Hydrolysis, decarboxylation, and the choice of base were discussed on p. 597.

Diethyl malonate adds to diethyl fumarate in a conjugate addition reaction promoted by sodium ethoxide in dry ethanol to give a tetraester. Diethyl fumarate is an excellent Michael acceptor because two ester groups withdraw electrons from the alkene. The mechanism involves deprotonation of the malonate, conjugate addition, and reprotonation of the product enolate by ethanol solvent. In this reaction two ester groups stabilize the enolate and two more promote conjugate addition.

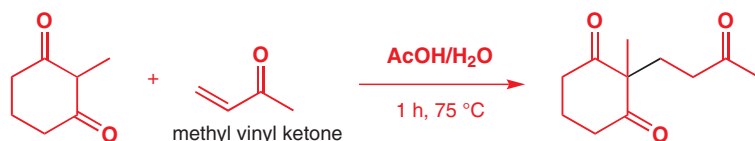


The value of malonate esters is illustrated in this synthesis of a substituted cyclic anhydride by conjugate addition to ethyl crotonate, hydrolysis, and decarboxylation, followed by dehydration with acetic anhydride. This route is very general and could be used to make a range of anhydrides with different substituents simply by choosing an appropriate unsaturated ester.



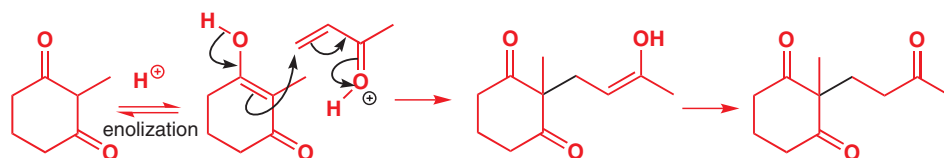
If the nucleophile is sufficiently enolized under the reaction conditions then the enol itself is able to attack the unsaturated carbonyl compound. Enols are neutral and thus soft nucleo-

philes favouring conjugate attack. 1,3-Diketones are enolized to a significant extent (Chapter 20), and under acidic conditions conjugate addition proceeds very efficiently even though there can be absolutely no base present. In this example methyl vinyl ketone (butenone) reacts with a cyclic β -diketone promoted by acetic acid to form a quaternary centre.

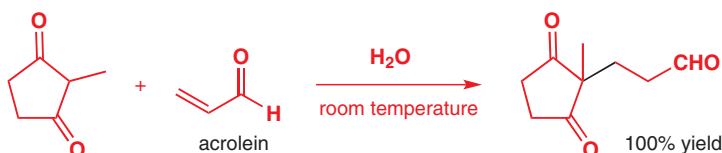


➡ The triketone product is an important intermediate in steroid synthesis as you will see in Chapter 26, p. 652.

The mechanism involves acid-catalysed conversion of the keto form of the cyclic β -diketone into the enol form, which is able to attack the protonated enone. The mechanistic detail is precisely analogous to the attack of an enolate; the only difference is that both reactants are protonated. The product is the enol form of the triketone, which rapidly tautomerizes to the more stable keto form.



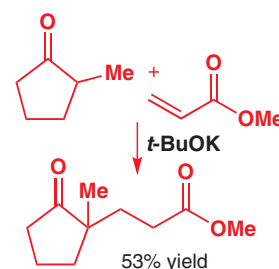
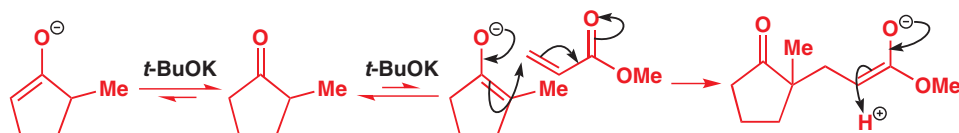
The thermodynamic control of conjugate addition allows even enals that are very electrophilic at the carbonyl carbon to participate successfully. As you will see in the next chapter, an aldol reaction (direct addition to C=O) must be possible here, but it is reversible and 1,4-addition eventually wins out. Acrolein combines with this five-membered diketone under very mild conditions to give a quantitative yield of product.



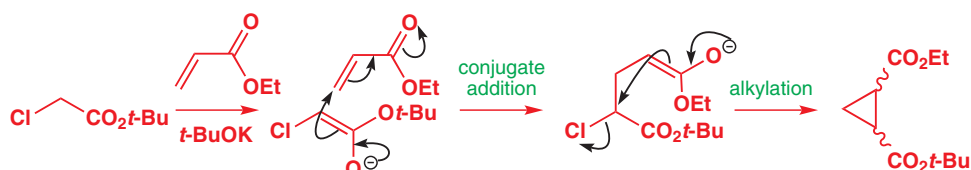
Alkali metal (especially Na, K) enolates can undergo conjugate addition

The use of two anion-stabilizing groups is a sure way of promoting conjugate addition, but it is not essential. Simple lithium enolates are not ideal nucleophiles for thermodynamically controlled conjugate addition because lithium binds strongly to oxygen and so tends to stabilize the aldol product. Better results are often observed with sodium or potassium enolates, which are more dissociated. Potassium *tert*-butoxide is the ideal base for this example as it is hindered and so will not attack the ester but is basic enough to deprotonate the ketone to a certain extent.

Two enolates are possible but, under the equilibrating conditions, the more stable enolate is the one leading to the product with a quaternary carbon atom.

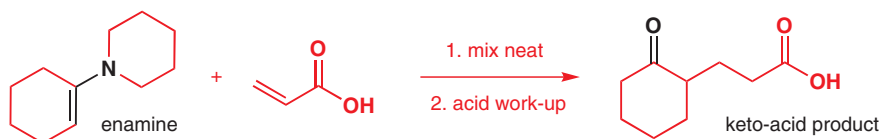


If the enolate carries a leaving group, we get a nice way of making a cyclopropane because the enolate formed by the conjugate addition can itself be alkylated.

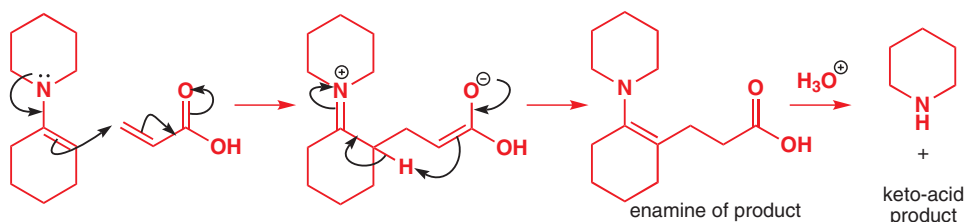


Enamines are convenient stable enol equivalents for conjugate addition

If you want a more reliable way of doing a conjugate addition of an aldehyde or ketone without having a second anion-stabilizing group, you need some stable and relatively unreactive enol equivalent. On p. 591 you saw how enamines, particularly those derived from cyclic secondary amines, are useful in alkylation reactions. These neutral species are also perfect for conjugate addition as they are soft nucleophiles but are more reactive than enols and can be prepared quantitatively in advance. The reactivity of enamines is such that heating the reactants together, sometimes neat, is all that is required. Acid catalysis can also be used to catalyze the reaction at lower temperature.

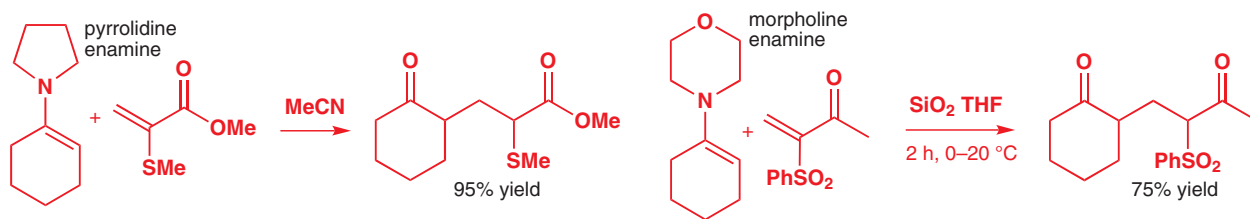


The mechanism is rather like enol addition. The differences are that the enamine is more nucleophilic because of the nitrogen atom and that the product is also an enamine, which can be converted into the corresponding carbonyl by mild acidic hydrolysis. This is usually performed during the work-up and so does not really constitute an extra step. The amine is washed out as the hydrochloride salt so isolation is straightforward. After conjugate addition the resulting enolate-iminium ion undergoes proton transfer rapidly to produce the more stable carbonyl-enamine tautomer. This is shown as an intramolecular process but it could just as easily be drawn with an external base and source of protons. The resulting enamine is then stable until aqueous acid is added at the end of the reaction. Hydrolysis occurs via the iminium ion to reveal the second carbonyl group and release the secondary amine.



➡ In Chapter 41 we will discuss the catalytic use of chiral enamines to promote related reactions in asymmetric form.

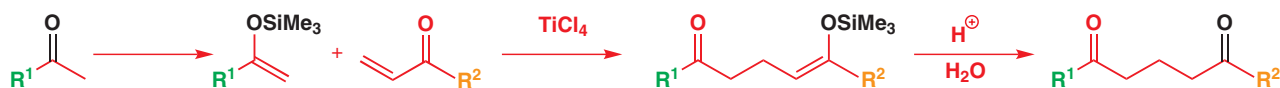
In these two examples enamines from cyclohexanone formed with pyrrolidine and morpholine add in good yield to an α,β -unsaturated carbonyl compound with an extra electron-withdrawing methylthio or phenylsulfonyl group.



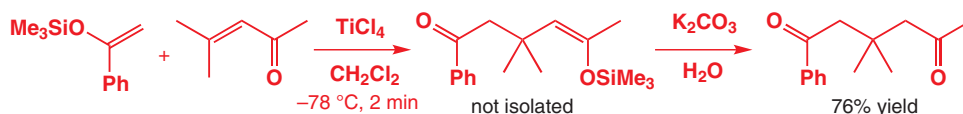
➡ See Chapter 20 for an introduction to silyl enol ethers and p. 466 for a description of Lewis acids. You saw a reaction similar to this with an alkyl halide as an electrophile on p. 604.

Conjugate addition of silyl enol ethers leads to the silyl enol ether of the product

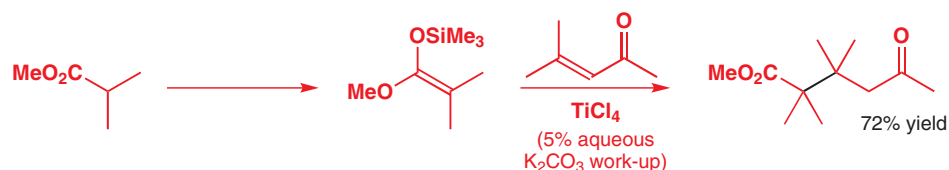
The best alternatives to enamines for conjugate addition of enols of aldehyde, ketone, and carboxylic acid derivatives are silyl enol ethers. These stable neutral nucleophiles react very well with Michael acceptors either spontaneously or with Lewis acid catalysts such as TiCl_4 at low temperature. If the 1,5-dicarbonyl compound is required, then an aqueous work-up with either acid or base cleaves the silicon-oxygen bond in the product.



Addition of the silyl enol ether derived from acetophenone (PhCOMe) to a disubstituted enone promoted by titanium tetrachloride is very rapid and gives the diketone product in good yield even though a quaternary carbon atom is created in the conjugate addition. This is a typical example of this very powerful class of conjugate addition reactions.

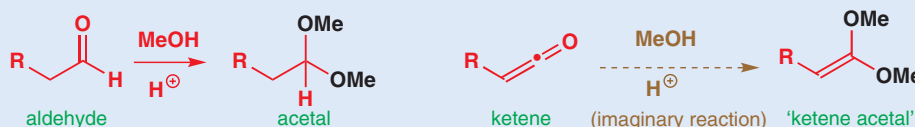


It is even possible to use a silyl enol ether to create a new C–C bond that joins two new quaternary centres. Silyl ketene acetals (the silyl enol ethers of esters) are more nucleophilic than ordinary silyl enol ethers, and in this example the silyl ketene acetal undergoes conjugate addition to an unsaturated ketone catalysed by the usual Lewis acid (TiCl₄) for such reactions.

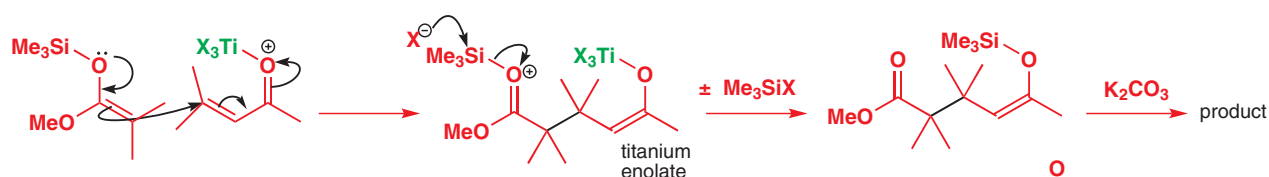


Ketene acetals

Because enol ethers of esters have two identical OR groups joined to the same end of the same double bond, you will see them called 'ketene acetals' or, here, 'silyl ketene acetals'. This is a reasonable description as you can imagine the carbonyl group of a ketene forming an acetal in the same way as an aldehyde. In fact, they cannot be made this way.

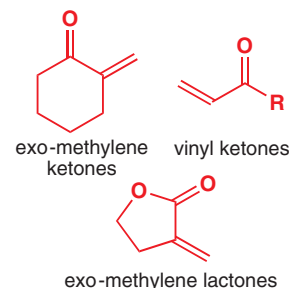


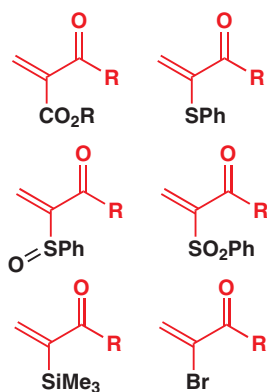
In these reactions, the electrophile coordinates to the TiCl₄ Lewis acid first, producing an activated enone that is attacked by the silylated nucleophile. It is difficult to determine at what stage the trimethylsilyl group moves from its original position and whether it is transferred intramolecularly to the product. In many cases the anion liberated from the Lewis acid (Cl[−], RO[−], Br[−]) is a good nucleophile for silicon so it is reasonable to assume that there is a free trimethylsilyl species (Me₃SiX) that captures the titanium enolate:



A variety of electrophilic alkenes will accept enol(ate) nucleophiles

The simplest and best Michael acceptors are those α,β -unsaturated carbonyl compounds with exposed unsaturated β carbon atoms, such as *exo*-methylene ketones, lactones, and vinyl ketones. However, their extreme reactivity can make them hard to handle (they polymerize readily), and in the next chapter (p. 621) you will meet a method that makes them *in situ* to circumvent these problems.

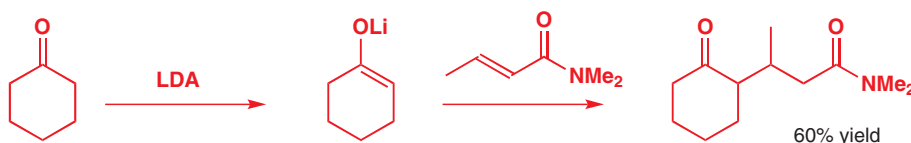




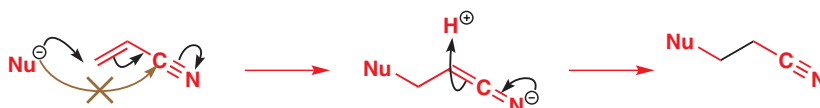
➡ The selective activation achieved by a nitrile group in enolate alkylation was explained at the start of this chapter.

One trick to persuade a stubborn enolate to do conjugate rather than direct substitution is to add an extra anion-stabilizing substituent in the α position. The margin shows a selection of reagents that do this. In each case the extra group (CO₂Et, SPh, SOPh, SO₂Ph, SiMe₃, and Br) can be removed after the conjugate addition is complete.

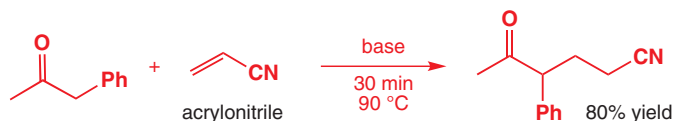
Unsaturated **esters** are good Michael acceptors because they are not very electrophilic. Unsaturated **amides** are even less electrophilic and (provided they are tertiary and have no acidic NH protons) will even give conjugate addition products with lithium enolates.



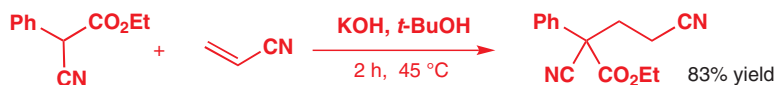
The **nitrile** group is not as reactive towards direct attack by nucleophiles as its carbonyl cousins but is equally able to stabilize an adjacent negative charge. Alkenes conjugated with nitriles are thus activated towards nucleophilic attack without the complications of competing direct addition to the activating group.



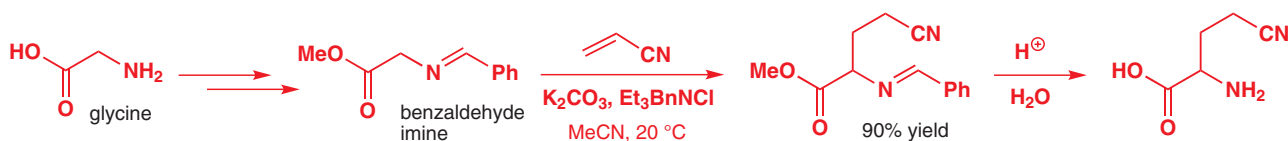
With base, methyl benzyl ketone forms its more stable enolate, which undergoes smooth and rapid conjugate addition to acrylonitrile. Acrylonitrile is one of the best Michael acceptors for enolates.



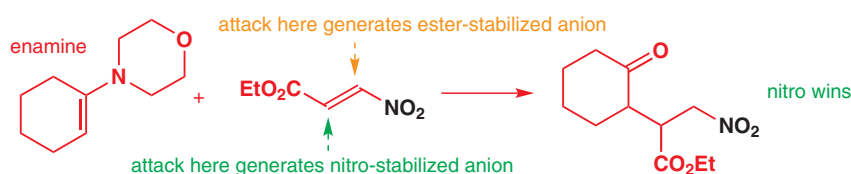
The cyanide group can also act as an anion-stabilizing group in the nucleophile. In combination with an ester group, the enolizable proton is acidified to such an extent that potassium hydroxide can be used as base.



The simplest amino acid, glycine, would be an ideal starting material for the synthesis of more complicated amino acids but it does not easily form enols or enolates. By conversion to the methyl ester of its benzaldehyde imine, two electron-withdrawing groups are introduced to help stabilization of the enolate and conjugate addition of acrylonitrile is now possible. The base used was solid potassium carbonate. Simple hydrolysis of the alkylated product leads to the extended amino acid.

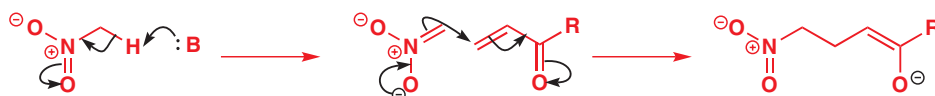


You saw on p. 606 how two ester groups in fumarate diesters encourage conjugate addition, but what if there are two *different* groups at the ends of the Michael acceptor? Then you must make a judgement as to which is more electron-withdrawing. One case is clear-cut. The **nitro** group is worth two carbonyl groups (p. 586) so that conjugate addition occurs β to the nitro group in this case.



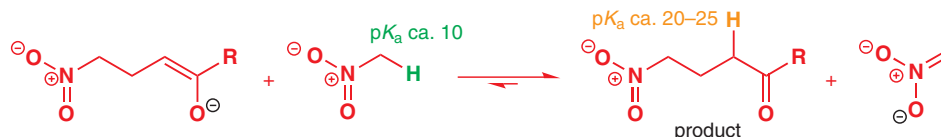
Nitroalkanes are superb nucleophiles for conjugate addition

In this chapter so far you have seen that highly stabilized anions, such as those derived from β -dicarbonyl compounds, are particularly good at nucleophilic addition because their stability helps to reverse the unwanted alternative direct $C=O$ addition (aldol) pathway, and facilitates proton transfer in the catalytic version of the reaction. The nitro group is so powerfully electron-withdrawing that just one is equivalent to two carbonyls in pK_a terms (p. 586). Thus if β -dicarbonyls are good for conjugate addition, you might expect nitroalkanes to undergo conjugate addition in just the same way. The good news is that they do, very well. The first stage is a base-catalysed conjugate addition.

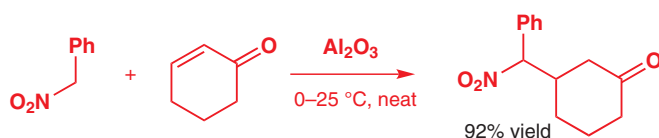


■ Always draw out the nitro group in full when using it in mechanisms.

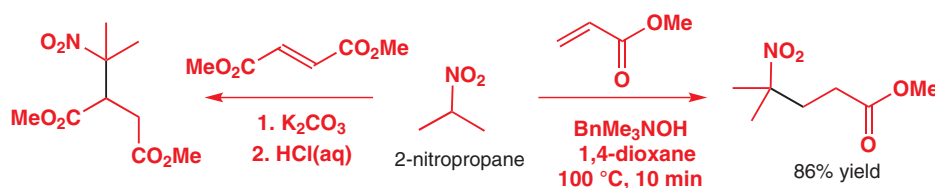
The product enolate that is formed is much more basic than the anion of the nitro compound so it removes a proton from the nitro compound and provides another molecule of anion for the second round of the reaction.



The acidifying effect of the nitro group is so profound that very mild bases can be used to catalyse the reaction. This enables selective removal of the proton next to the nitro group and helps to avoid side reactions of the carbonyl component. Common examples of mild bases include amines, quaternary ammonium hydroxides, and fluorides. Even basic alumina (a largely inert powder) is sufficient to catalyse virtually quantitative addition of this benzylic nitroalkane to cyclohexenone at room temperature!



Anions of nitro compounds form quaternary centres with ease in additions to α,β -unsaturated mono- and diesters. The difference between the acidity of the protons next to a nitro group and those next to the esters in the products combined with the very mild basic conditions ensures that no unwanted side reactions occur.

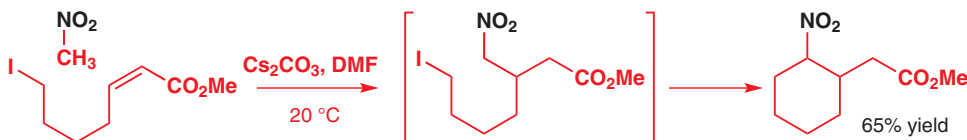


■ As you will see in Chapter 26, enolizable esters, ketones, and aldehydes are prone to undergo condensation reactions with themselves in the presence of strong bases.

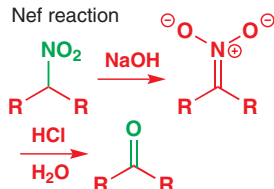
The effectiveness of nitro compound conjugate addition makes it ideal for use in combination with other reactions in making several bonds in one pot. The next example combines conjugate addition and intramolecular conjugate addition to make a six-membered ring.

■ Caesium is the most electropositive of readily available metals: electronegativity = 0.79.

The base used for both steps is Cs_2CO_3 . The large caesium cation forms fully ionic compounds so the uncomplexed carbonate ion can exert its full basicity. Deprotonation of the conjugate addition product next to the nitro group produces a second anion, which does an intramolecular $\text{S}_{\text{N}}2$ displacement of iodide to form a six-membered ring.

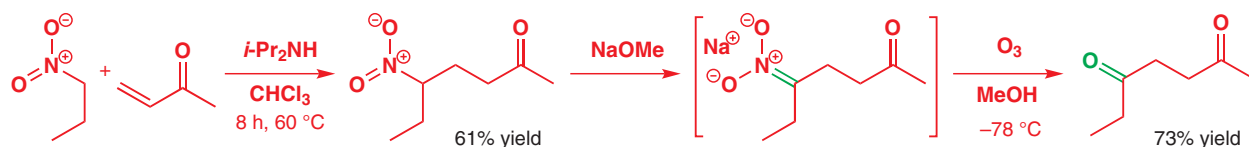


Nef reaction



➡ Ozonolysis was described in Chapter 19.

The nitro group can be converted into other useful functional groups following conjugate addition. Reduction gives primary amines while hydrolysis reveals ketones. The hydrolysis is known as the **Nef reaction** and used to be achieved by formation of the nitro-stabilized anion with a base such as sodium hydroxide followed by hydrolysis with sulfuric acid. These conditions are rather unforgiving for many substrates (and products) so milder methods have been developed. One of these involves reaction of the nitro 'enolate' with ozone (ozonolysis) at low temperature rather than treatment with acid. Base-catalysed conjugate addition of nitropropane to methyl vinyl ketone occurred smoothly to give the nitroketone. Formation of the salt with sodium methoxide was followed by oxidative cleavage of the $\text{C}=\text{N}$ linkage with ozone. The product was a 1,4-diketone, which was isolated without further aldol reaction by this route.

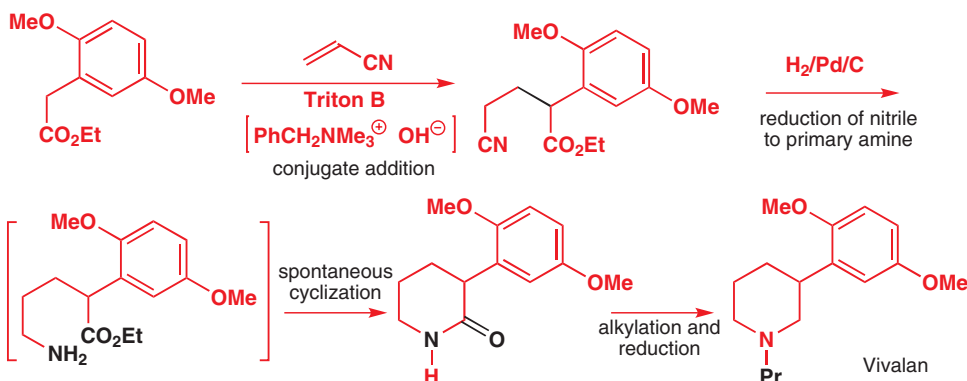


This is a good general method for the synthesis of 1,4-diketones, which can otherwise be difficult to make, and additional substituents are easily accommodated on the enone—a characteristic of conjugate addition.

The synthesis of a drug that acts on brain chemistry

We end this chapter with a simple commercial synthesis of a drug molecule—vivalan—described as a 'dopaminergic antagonist'. It uses four reactions that you have met: conjugate addition of an enolate to acrylonitrile, reduction of CN to a primary amine, alkylation, and reduction of the amide. There is another reaction involved—cyclization to an amide—but this occurs spontaneously.

■ The conjugate addition uses a base, benzyltrimethylammonium hydroxide, marketed as Triton B, which allows hydroxide to be soluble in organic solvents.



To conclude...

We have considered the reactions of enolates and their equivalents with alkyl halides and electrophilic alkenes. In the next chapter we move on to consider reactions we have deliber-

ately taken steps to avoid up to this point. We shall consider the same types of enolate equivalents reacting with carbonyl compounds themselves.

● Summary of methods for alkylating enolates

Specific enol equivalent	Notes
<i>To alkylate esters</i>	
- LDA → lithium enolate - use diethyl- or dimethylmalonate and decarboxylate	gives acid (NaOH, HCl) or ester (NaCl, DMSO)
<i>To alkylate aldehydes</i>	
- use enamine - use silyl enol ether - use aza-enolate	with reactive alkylating agents with S _N 1-reactive alkylating agents with S _N 2-reactive alkylating agents
<i>To alkylate symmetrical ketones</i>	
- LDA → lithium enolate - use acetoacetate and decarboxylate - use enamine - use silyl enol ether - use aza-enolate	equivalent to alkylating acetone with reactive alkylating agents with S _N 1-reactive alkylating agents with S _N 2-reactive alkylating agents
<i>To alkylate unsymmetrical ketones on more substituted side</i>	
- Me₃SiCl, Et₃N → silyl enol ether - Me₃SiCl, Et₃N → silyl enol ether → lithium enolate with MeLi - alkylate acetoacetate twice and decarboxylate - addition or reduction of enone to give specific lithium enolate or silyl enol ether	with S _N 1-reactive alkylating agents with S _N 2-reactive alkylating agents two successive alkylations of ethyl acetoacetate
<i>To alkylate unsymmetrical ketones on less substituted side</i>	
- LDA → kinetic lithium enolate - LDA then Me₃SiCl → silyl enol ether - use dianion of alkylated acetoacetate and decarboxylate - use enamine	with S _N 2-reactive electrophiles with S _N 1-reactive electrophiles two successive alkylations of ethyl acetoacetate with reactive electrophiles

Further reading

P. Wyatt and S. Warren, *Organic Synthesis: Strategy and Control*, Wiley, Chichester, 2007, chapter 10. An early article by a pioneer in this field is H.O. House, M. Gall, and H. D. Olmstead, *J. Org. Chem.*, 1971, **36**, 2361. A good example of specific enolate forma-

tion and alkylation from an enone is D. Caine, S. T. Chao, and H. A. Smith, *Org. Synth.*, 1977, **56**, 52. F. A. Carey and R. J. Sundberg, *Advanced Organic Chemistry B, Reactions and Synthesis*, 5th edn, Springer 2007, chapter 1.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>